

MERGER AND SCATTERING OF TRAVERSABLE WORMHOLES: GRAVITATIONAL RADIATION FROM UNSTABLE BINARIES

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(Dated: September 21, 2026)

We evolve pairs of massive drainhole wormholes—phantom-scalar throats—in full numerical relativity and map the outcomes available to a pair at one throat compactness, equal masses and no spin. A lone throat is an unstable fixed point (e-fold time $\tau = 5.9$, within 15% of the González-Guzmán-Sarbach rate) with two branches, collapse and inflation; a declared kick selects the branch deterministically and accelerates it. Like-oriented throats repel, so a merger requires opposite scalar signs. A head-on pair then collides with both mouths still open and forms a black hole, the trapped surface born enclosing both mouths and never either alone, which loses mass to the phantom field it swallows ($M_{\text{MS}} 3.0 \rightarrow 2.2$) and rings without a chirp. The spiral merger instead stalls: no horizon exists anywhere on its measured record, the curvature wall that kills every grid rides the still-open throat, and a $16\times$ refinement ladder saturates without reaching the merger’s end. An e-fold budget built on three agreeing measurements of the unstable mode forbids a wormhole inspiral outright—an orbit costs $(d/M)^4$ while a throat endures only $\ln(1/\varepsilon)$, so surviving one orbit from the innermost stable circular orbit needs a seed below 10^{-20} . Every encounter is therefore a plunge, and the loudest gravitational channel is the horizonless fly-by, radiating $7.4 \times 10^{-2} M$ (43% spread over extraction spheres) in the dominant multipole— $29\times$ this campaign’s vacuum merger control and, against a vacuum pass carrying the fly-by’s own momentum and run for this comparison, $70\times$: the momentum-matched control is itself $2.3\times$ quieter than the merger control, and that factor is what the older number was missing. The gravitational channel is not the whole emission: on the same extraction sphere the ghost scalar carries 2.3 times the gravitational energy, in the $\ell = 1$ dipole that a vacuum binary has no analogue for, and—because gravity couples to minus this field’s stress tensor—with the opposite sign, so the system is pumped by its own dominant radiation rather than damped by it. Rescaled to stellar mass and carried through the detector data products, these bursts are not separated from binary black holes by morphology at fixed duration (fitting factors 0.82–0.98 against a vacuum control’s 0.94–0.95); the discriminator is the frequency track, which falls or drifts where a remnant’s would hold at the Kerr value. A matched-filter search of public Advanced LIGO O3b strain with a 127-template bank returns no candidate at zero lag, at a measured single-detector reach of 0.15–4.8 Gpc across the channels.

I. INTRODUCTION

Wheeler’s spacetime foam is a sea of handles—wormholes that open and close on the Planck scale, both mouths in the same space [1–3]. More than one mechanism promotes them to macroscopic size: a Planck-scale Lorentzian wormhole caught in an inflating background is stretched with it and emerges macroscopic [4]; bubble and domain-wall cosmologies produce, without tuning, post-inflationary regions joined through wormholes [5, 6]; and a gas of relic wormholes has been run as a cosmology in its own right [7, 8]. What none of this work evolves is the aftermath: what a relic wormhole does once it is macroscopic and classical, and what two of them do to each other. That aftermath is this paper’s subject.

The classical theory fixes its two clocks. First, the simplest evolvable traversable wormhole—the Ellis-Bronnikov drainhole held open by a ghost scalar [9–12]—is unstable: perturbed one way the throat collapses to a black hole, the other way it inflates, on a timescale set by the throat radius [13, 14], as a 3+1 evolution of

the massless throat confirms nonlinearly [42]. Every constituent destroys itself on its own schedule. Second, Geroch’s theorem forbids classical topology change [15, 16]: two wormholes cannot fuse into one, and none can be born. Birth is quantum [17], so a simulation takes over at $t = 0$ with the newborn state as its initial data—the stipulation every numerical-relativity binary makes, here with a physical story attached.

Together the two facts turn the standing objection to wormhole binaries into a selection rule. A binary has no astrophysical assembly channel—each throat dies faster than capture could pair two—but a foam-born pair needs none: creation and collision are the same event, at a birth separation of order the objects themselves. The race is collision against decay, and both rates belong to the classical solution. The scalar’s sign adds a second rule at no cost: a phantom field repels like charges and attracts opposite ones, so like-signed pairs scatter while opposite-signed pairs fall together; and the sign of a lone throat’s perturbation selects collapse—a wormhole-to-primordial-black-hole channel beside the standard routes [18–20]—or inflation. A foam population thus fans out into four fates—merge, scatter, collapse, inflate—across the three-parameter space of birth separation, perturbation and relative sign, and that space is what a family of evolu-

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tions maps. The binary evolved here is the exterior approximation of a pair of handles (three asymptotic ends, the far sides passive); what merges is not two wormholes into one, which Geroch forbids, but two wormholes into one black hole with both throats inside it—the horizon does the work that topology change cannot.

The observable is not a chirp but a stochastic background from the merger epoch, redshifted to whichever band the mass scale puts it in [21–24]. The clean deliverable is therefore dimensionless and epoch-independent: the energy radiated per merger and its spectral shape in throat-mass units, which converts any bound on Ω_{GW} —including the nucleosynthesis cap that holds outside every detector band [25]—into a bound on the co-moving abundance of primordial wormhole mergers. Two caveats travel with the statement: the ghost scalar must exist at that epoch to hold the newborn throats open, and the merger’s scalar channel—measured here at 2.3 times the gravitational energy and negative (Sec. VIII F)—has a cosmological consequence that is genuinely open.

The paper is organised as follows. Sec. II sets out the model: the Einstein–phantom-scalar system, the drainhole, the binary superposition with its sign rule, and the declared seeds. Sec. III gives the numerical setup and the run matrix. The results then follow the campaign from one throat to two: the lone wormhole—fixed point, unstable mode, branching and collapse (Sec. IV); two throats at rest and the force between them (Sec. V); the head-on merger to a black hole (Sec. VI); the spiral merger and its capture boundary (Sec. VII); the gravitational waves of every channel—collapse, head-on, spiral, fly-by—gathered in one chapter with the vacuum twin (Sec. VIII); the numerical controls and systematics (Sec. IX). Sec. X confronts the rescaled waveforms with the LIGO–Virgo–KAGRA data products—template banks, matched-filter search, ring-down, burst searches—and Sec. XI collects the findings, including the e-fold budget that forbids a wormhole in-spiral (Sec. XI A) and the list of what is not established (Sec. XI B). Secs. XII and XIII give the reproducibility pack and the computational cost. Every evolution runs on GPUs with GRTeclyn [39], the AMReX-based [40] port of GRChombo [41], continuing the campaign of Refs. [42, 43]. Throughout, $G = c = 1$.

II. THE MODEL

A. Field equations

The matter is a single real, massless phantom scalar: gravity couples to *minus* the canonical stress tensor,

$$T_{ab} = -\left(\nabla_a \phi \nabla_b \phi - \frac{1}{2} g_{ab} \nabla_c \phi \nabla^c \phi\right), \quad V(\phi) \equiv 0, \quad (1)$$

so $\rho < 0$ where the field’s gradient dominates—the exotic stress that holds a throat open. With $V = 0$ this is exactly a ghost field. The sign flip lives entirely in the

coupling to geometry: the scalar itself obeys the canonical wave equation $\square \phi = 0$, evolved first order in ϕ and $\Pi = (\partial_t \phi - \beta^i \partial_i \phi)/\alpha$.

B. Drainhole initial data

Each body is a massive Ellis–Bronnikov drainhole [9, 10]. In isotropic radius r about the throat centre, with

$$X = \frac{r - a^2/4r}{a}, \quad \Omega = 1 + \frac{a^2}{4r^2}, \quad u = \frac{m}{a} \left(\arctan X - \frac{\pi}{2} \right), \quad (2)$$

the exact static solution of Eq. (1) is

$$\alpha = e^u, \quad \gamma_{ij} = e^{-2u} \Omega^2 \delta_{ij}, \quad \phi = C \arctan X, \quad (3)$$

with $4\pi a^2 C^2 = a^2 + m^2$ (so $C \rightarrow 1/\sqrt{4\pi}$ as $m \rightarrow 0$), $K_{ij} = 0$, $\Pi = 0$ and zero shift; in the evolved conformal variables $\chi = e^{2u} \Omega^{-2}$ and $h_{ij} = \delta_{ij}$. Both constraints hold identically at $t = 0$. Here a is the throat scale and m the ADM mass, carried by the *lapse* rather than the conformal factor, so χ at the throat stays in $[0.15, 0.25]$ for $m/a \leq 1$ and mass costs no grid resolution. The minimal surface sits at $r_t = \frac{1}{2}(m + \sqrt{m^2 + a^2})$ with areal radius $R_{\min} = e^{-u(r_t)} \sqrt{a^2 + m^2}$; the production throat $a = 2$, $m = 1$ has $R_{\min} = 3.8895$ at $r_t = 1.618$. The point $r \rightarrow 0$ is the far universe’s spatial infinity compactified ($\chi \sim r^4$). The mass matters: at $m = 0$ the throat is ultrastatic ($\alpha \equiv 1$, $M_{\text{ADM}} = 0$) and two of them do not attract at all.

C. The binary superposition

Two throats at separation d (one-body radii r_A , r_B) are superposed in excess form,

$$u = u_A + u_B, \quad \psi = 1 + (\sqrt{\Omega_A} - 1) + (\sqrt{\Omega_B} - 1), \quad (4)$$

with $\chi = e^{2u} \psi^{-4}$, $\alpha = e^u$ and $\phi = \phi_A + \sigma \phi_B$, $\sigma = \pm 1$; the constant tail of the arctan profiles is subtracted so $\phi \rightarrow 0$ at the boundary, which is free for a massless field. Momentum enters through one Bowen–York term [33] per throat,

$$\hat{A}_{ij} = \frac{3}{2r^2} [P_i n_j + P_j n_i - (\delta_{ij} - n_i n_j) P \cdot n], \quad (5)$$

converted by $\tilde{A}_{ij} = \chi^{3/2} \hat{A}_{ij}$, with $K = 0$. Since $\Pi = 0$ the matter momentum density vanishes and $\hat{A}_{ij}$ is flat-space divergence free, so the momentum constraint is satisfied *exactly* at $t = 0$; only the Hamiltonian constraint carries the superposition defect, $O(a^2/d^2) + O(m/d)$. The relative sign σ is the pair’s one discrete parameter: $\phi \rightarrow -\phi$ is exact for a single body, but a phantom mediator *repels* like scalar charges and *attracts* opposite ones, with $|F_\phi/F_{\text{grav}}| = (a^2 + m^2)/m^2 > 1$ always ($= 5$ for the production throat)—so like-signed pairs can never merge under their own gravity, and $\sigma = -1$ is the merger channel.

A windowed one-body correction of the Helfer type [34] is implemented but off in production (Sec. IX A).

D. Declared seeds

The instability is probed with declared perturbations of the conformal factor only: a Gaussian shell on the minimal surface,

$$\psi \rightarrow \psi[1 + \varepsilon g(r)], \quad g = e^{-(r-r_t)^2/w^2}, \quad w = a/4, \quad (6)$$

which moves the throat's areal radius by $\approx 2\varepsilon$ with the sign of ε , and a quadrupolar factor

$$\psi \rightarrow \psi[1 + \varepsilon_2 g(r) P_2(\cos \theta)] \quad (7)$$

about the z axis through the throat. K_{ij} and Π are untouched, so the momentum constraint stays exact. The spherical seed selects the branch; the quadrupolar seed has no $\ell = 0$ part—it gives a collapsing throat a quadrupole to shed—and with both on the cross term is $O(\varepsilon\varepsilon_2)$.

III. NUMERICAL SETUP

The campaign is one-knob: every run differs from a named reference in a single setting, and every run has one line—its difference, its outcome, its caveats—in a registry shipped with the results pack (Sec. XII). Table I groups the 132 evolutions; the verdicts are the business of Secs. IV–IX. An archived shakedown stage (11 of the 132) fixed the gauge, dissipation and tagging choices below and is not used for physics, which leaves the 121 evolutions the results rest on (Sec. XIII).

A. Evolution scheme

Evolutions use GRTeclyn: CCZ4 [35] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, $1 + \log$ slicing [36] started from the exact static lapse $\alpha = e^u$ of Eq. (3), a Gamma-driver shift [37, 38] with $\eta = 1$ started from zero, fourth-order stencils, RK4, and Kreiss–Oliger dissipation $\sigma = 0.1$. The time step is $\Delta t = 0.02 \Delta x$, an order of magnitude below the usual CCZ4 Courant factor; the bracket in Δt that makes it necessary is a control (Sec. IX A).

B. Grid, boundaries and regularisation

The domain is a cube of side $L = 64$ with the system at its centre; the base grid is 128^3 ($\Delta x = 0.5$). Refinement is by moving nested boxes centred on the tracked throats, shrinking by a factor of two per level (half-width 4 per throat at the production finest level) and re-drawn

every 16 steps. Production runs use three refinement levels (finest $\Delta x = 1/16$); resolution arms raise this to levels 4–7 where stated, and the orbital production chain and the fly-by double the box to $L = 128$, carrying the sponge and the extraction spheres outward with it. Outer boundaries are Sommerfeld, backed by an absorbing sponge over $r = 24$ –32 that ramps extra dissipation up to $\sigma_{\text{sp}} = 4$ as the fourth power of the depth into the layer; everything dynamical sits inside $r = 24$.

Each throat carries a compactified second infinity at its own centre, where $\chi \sim r^4$, so the evolved variables are floored: $\chi \geq 10^{-8}$ and $\alpha \geq 10^{-10}$ (a variant keeps χ unfloored down to 10^{-20} and applies the 10^{-8} floor only where $1/\chi$ is evaluated). Two further devices act only inside regions that are already trapped: an exponential damping of the scalar where the lapse has collapsed, and a smooth interior freeze that multiplies every right-hand side by $1 - W(r)$, with W a C^2 quintic step between an inner radius r_{full} and an outer r_{start} . All three are numerical devices, not physics; whether any of them is load-bearing is itself measured by twin runs (Secs. VI B and IX A).

C. Diagnostics

L^2 norms of the Hamiltonian and momentum constraints over the box are written continuously. Both throat centres are tracked, and about each centre the areal radius $R(r)$ is scanned on coordinate spheres; its minimum is the throat radius. On the same spheres the two null expansions $\theta_{\pm}$ are computed with the outward direction fixed by *increasing areal radius*: a marginally outer trapped surface is an outermost zero of θ_+ with $\theta_- < 0$ (crossings at extrema of $R(r)$ are rejected), while a dynamic wormhole throat is marginally *anti*-trapped [32]. That orientation is not a refinement but a correction: the in-code $\theta_{\pm}$ stream fixes the outward direction by increasing *coordinate* radius, which on the far side of a throat points the wrong way and manufactures trapped verdicts wherever $R(r)$ is falling. Every horizon statement in this paper—formation, absence, radius, mass—is therefore read from an offline scan over the archived plotfiles with the areal orientation applied, and the in-code columns are used only as a live monitor and are never plotted. The correction is not cosmetic: on the naive orientation the spiral remnant reports a common horizon that the corrected scan does not find (Sec. VII B). Masses of such surfaces are Misner–Sharp [31],

$$\frac{2M_{\text{MS}}}{R} = 1 + \frac{R^2 \theta_+ \theta_-}{4}, \quad (8)$$

with $\theta_{\pm}$ carrying dimensions of inverse length so that the second term is dimensionless; the scan uses coordinate spheres rather than a deformable finder, and quoted masses—and the presence or absence of a surface, not only its mass—carry that shape systematic. Radiation is read from Ψ_4 interpolated onto spheres at $R = 10$,

TABLE I. The run matrix: 132 evolutions, grouped. Every group varies one knob against a named reference; per-run lines live in the shipped registry. Body parameters are $a = 2$, $m = 1$ ($R_{\min} = 3.89$), $\sigma = -1$, $d = 12$ and $L = 64$ unless stated. Left: the single throat, the pair and the head-on, with the controls. Right: the orbital family.

group	knob(s)	runs	Sec.	group	knob(s)	runs	Sec.
lone throat	levels 2–4; χ floor; Δt ; $L=128$	9	IV	capture scan	$p = 0.20\text{--}0.45$	4	VII A
spherical kicks	$\varepsilon = \pm 10^{-3} \dots \pm 10^{-1}$; level 4	9	IV C	fly-by, $p = 0.45$	level 5 on the doubled box	2	VIII D
quadrupolar kicks	$\varepsilon_2 = 0.005\text{--}0.05$; kick-free arm	6	VIII A	orbital chain, $p = 0.12$	gauge; N ; levels; damping; χ floor	16	VII
two throats at rest	σ ; $a = 1\text{--}3$; $d = 12\text{--}18$	9	V	refinement ladder	levels 3–7; Δt ; damping	8	VII C
placement probes	$d = 6\text{--}48$, one step each	18	V C	interior freeze, $L=64$	skin radii; arming time; extraction	7	VIII C
head-on, $d = 8$	fill; levels 3/5; down-step; $\pm\varepsilon$	10	VI	wall probes	$p = 0.15\text{--}0.25$; levels; fill off	6	VII C
half-mass, $m = 0.5$	pair at $d = 6$; lone control	2	VIC	production spiral, $L=128$	three-stage chain; fill-window twin	6	VII B
vacuum BBH	same d , p , masses; no scalar	2	VIII E	Helfer twins	correction; window; levels	6	IX A
constraint-solved data	GRTresna bridge	1	IX A	shakedown (archived)	gauge, dissipation, tagging	11	III A

14, 18 (and 22 on the single-throat wave arms) on the $L = 64$ box, and at $R = 20, 28, 36, 44$ on the doubled one, decomposed in spin-weight -2 harmonics. *Every* amplitude in this paper, in text and in every figure, is $r\Psi_4$: both pipelines multiply by the sphere radius before the harmonic projection, so neither stream is a bare Ψ_4 that a reader—or a plotting script—should rescale. The two agree to 0.1% where they overlap, which is what makes the convention checkable rather than declared. The same spheres also carry a scalar stream: ϕ and Π projected on spin-weight-0 harmonics with the same radial scaling, $A_{\ell m} = R\phi_{\ell m}$, and the kinematic flux $F_{\text{kin}} = -R^2 \oint \Pi \partial_r \phi d\Omega$ signed so that an outgoing *canonical* wave gives a positive number (Sec. VIII F). A per-cell autopsy reports the location and one-step history of any floating-point failure.

IV. THE SINGLE WORMHOLE

A. The static throat as a fixed point

Evolved with no declared perturbation, the exact drainhole holds its areal throat radius $R_\star = 3.8895$ to better than 0.1% for 35 time units at level 3 and to 1% until $t \simeq 44$ (Fig. 1(a)); at level 4 the radius is still exact to four digits at $t = 30$. The static throat is a fixed point of the discrete system as well as of the continuum one. What ends the quiet phase is not a failure of the data: the departure time moves *later* by 5.6–8.7 units per halving of the cell, the signature of a truncation-noise seed of fixed growth rate being handed a smaller initial amplitude. Level 2 never reaches the branch at all—it dies of an origin failure at $t = 24.2$ with the throat still exact to 0.12%.

B. The unstable mode

The departure is a single exponential mode (Fig. 1(a,b)): plateau fits give e-fold times $\tau = 5.88$ (level 3) and 5.26 (level 4)—the same mode to 11%—at

$T = \tau \alpha_{\text{th}}/R_\star = 0.87$ and 0.78. González, Guzmán and Sarbach’s perturbative growth times for the Ellis throat run from $T = 0.846$ at their massless end to 0.590 in the opposite limit of their one-parameter family [14]; the level-3 value sits 3% above that massless endpoint and the level-4 value 8% below it. We do not claim a mass-resolved comparison: the map from this paper’s (a, m) to their parameters has not been checked term by term, so what is established is agreement with the published rate to $\sim 15\%$, not placement within a band (Sec. XIB). The instability is the throat’s; only its seed is the grid’s, and the seed’s *sign* flips with resolution: level 3 collapses, level 4 inflates. This confirms, from exact massive-drainhole data, the unstable behaviour and the two-branch structure found for the massless Ellis–Bronnikov throat in Ref. [42], where ambient noise drove inflation and a compressive perturbation drove collapse: the branches are the wormhole’s, not an artefact of either code.

C. Branching: collapse versus inflation

A declared kick removes the grid’s vote (Fig. 1(c)). At $\varepsilon = \pm 10^{-2}$ and $\pm 10^{-3}$ the fate is always opposite to the push, each $\pm$ pair crosses itself at the ε -independent $t_\times \simeq 13.0$, and a tenfold smaller kick arrives later by a fixed delay of 11–12 units, read at matched horizon radius; markers as far apart as the twins’ separation and first detection bracket it at 8–14, and the quoted figure is the matched-radius one throughout—linear response riding the same unstable mode. The branching survives refinement: at level 4, the grid whose own truncation seed *inflates* the unkicked throat, $+10^{-2}$ still forms its horizon at $t = 11$ (areal 3.883 against level 3’s 3.88) and -10^{-2} still inflates, the level-4 twins crossing at $t = 12.99$ against level 3’s 13.01. The branch point is thus fixed to 0.2% against both knobs—13.01 at $\varepsilon = 10^{-2}$ against 13.04 at 10^{-3} , and level 4 against level 3—which is what makes it the throat’s number rather than the grid’s or the kick’s. The kick also sets the clock: the collapse branch forms its horizon at $t = 11$ for $\varepsilon = +10^{-2}$ and

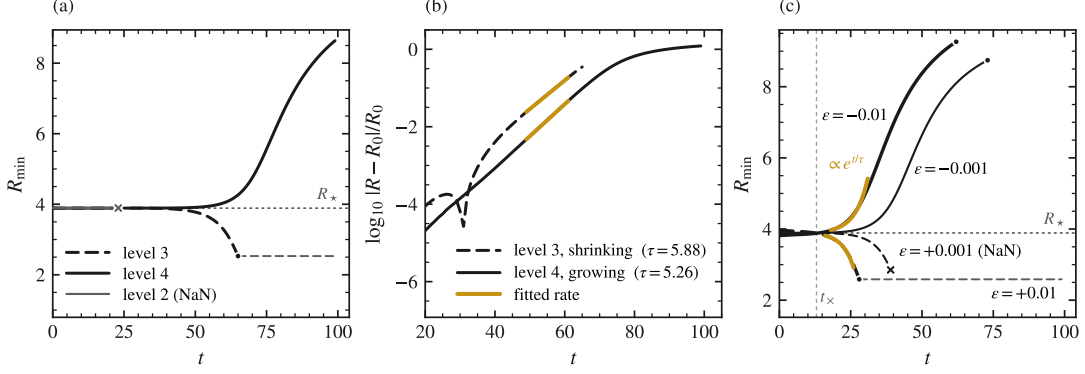


FIG. 1. The lone throat is unstable; unless the seed is declared, the grid picks the branch. (a) Minimal areal radius from exact static data at refinement levels 2–4: the exact solution is a fixed point (R_* , dotted), held until each grid’s own truncation noise excites the unstable mode—level 3 (dashed) collapses, level 4 (solid) inflates, level 2 (grey) dies of an origin failure first (cross). Dots mark where the *scan* loses the throat, not where the run ends; muted lines carry the last reading (the level-3 run continues to $t = 100$ behind a trapped surface). (b) The deviation is a single exponential mode with a level-dependent sign: plateau fits give $\tau = 5.88/5.26$ at levels 3/4, i.e. $T = 0.87/0.78$, within 15% of the González–Guzmán–Sarbach rate [14]. (c) Level 3 with the seed *declared*, Eq. (6): $\epsilon = \pm 10^{-2}$ (heavy), $\pm 10^{-3}$ (light); dashed = outward. The fate is opposite to the push, the twins cross at the ϵ -independent $t_\times \simeq 13.0$, and a tenfold smaller kick arrives 11–12 units later at matched horizon radius (8–14 across markers). Gold: exponential fits $|R - R_*| \propto e^{t/\tau}$; cross: death at $t = 40.1$.

$t = 25$ for $+10^{-3}$, against $t \simeq 61$ for the unkicked throat left to truncation noise—a driven branching an order of magnitude faster than the undeclared one. The dial has a ceiling: $\epsilon = \pm 0.1$ is not a perturbation—both signs collapse and die ($t = 15.2/14.1$) with a Hamiltonian violation 3–5 $\times$ the 1% arms’ and growing—so the usable amplitude ends at 10^{-2} .

D. The collapse endpoint

The collapse branch ends in a black hole that eats its own support: behind the marginally trapped surface the areal radius falls $3.23 \rightarrow 2.37$ and the Misner–Sharp mass $1.62 \rightarrow 1.23$ over 40 units—phantom infall *removes* mass—with the origin lapse at 0.016. (The massless throat of Ref. [42] bounced at $t \approx 4M$ and shed its horizon by $18.5M$; no arm here ever sheds a horizon, and the partial re-expansion below is a different thing.) The kicked $\epsilon = +10^{-2}$ arm runs clean to $t = 100$ behind its horizon, and the pure-quadrupole arm gives the endpoint its full instrumentation (Fig. 2): a MOTS from $t = 33$ at areal radius 3.80 settling to 2.41 with $M_{\text{MS}} 1.90 \rightarrow 1.24$, while the radial shells watch $|K|$ grow inward and the lapse pinch to zero at the centre—the first single-throat collapse recorded by the in-code radial profile from beginning to end.

The shrinking does not, however, continue. In the three kicked arms, which reach the floor 45 units earlier than the pure-quadrupole one and so have a record beyond it, the horizon bottoms out and then *grows back*: areal radius $2.00 \rightarrow 2.34$ (+17.0%, floor at $t = 48$) in the undamped $\epsilon = +10^{-2}$ arm, $2.27 \rightarrow 2.48$ (+9.4%, floor at $t = 47$) and $2.11 \rightarrow 2.42$ (+15.1%, floor at $t = 43$) in the

two scalar-damped ones, with M_{MS} recovering +10.6 to +12.5% in all three and the curves smooth and saturating rather than turning over. Three arms differing in the damping device agree, so the turnaround is not the device. The interior is meanwhile quasi-static—min α still creeping down, $10^{-2} \rightarrow 10^{-3}$, max $|K|$ flat at 0.23, min χ flat—so this is a horizon gaining mass around a frozen core, and it is the local counterpart of the negative-energy flux measured at infinity in Sec. VIII F: the remnant first loses mass to the phantom field it swallows and then regains a tenth of it. The pure-quadrupole arm of Fig. 2 reaches its own floor only at $t = 92$ and therefore shows no turnaround, which is the whole of why this record has to be read across arms. The inflation branch forms no horizon at all, and with the kick declared it can be watched to the end of a record rather than inferred from snapshots. A $\epsilon = -10^{-2}$ arm at level 4 carrying the collapse arms’ full instrument set (Fig. 3) grows $3.81 \rightarrow 11.39$ in areal radius, a factor 3.0, by $t = 100$ with no marginally trapped surface anywhere about the throat; what it does carry is that surface’s mirror, an anti-trapped shell ($\theta_+ > 0$ and $\theta_- > 0$) present at every scan centre from $t = 1$ to 35—the positive signature for which the earlier arms could report only an absence. Measured against its unkicked level-4 twin, the grid whose own truncation noise inflates the throat to +122%, the kicked arm departs by 10% at $t = 25$, the time by which the $+10^{-2}$ twin on this same level has already formed its horizon: the two signs leave together and end apart, and the departure rides the same mode ($\tau = 5.47$ from the plateau fit, between the levels’ 5.88 and 5.26). It also does not stay exponential. The *deviation* grows at the mode’s own 0.190 ± 0.002 per unit while it is small, but once the radius itself has doubled the growth decays—in the unkicked arm $d \ln R/dt$

TABLE II. The drainhole clock in physical units. The production throat ($a = 2$, $m = 1$, $R_* = 3.89 M$) scaled by its ADM mass M : $\tau = 5.9 M$ is the e-fold time of Sec. IV B, and the last three columns are the measured times to horizon formation for the two declared kicks and for the unkicked throat left to truncation noise (Sec. IV C). Intermediate seeds follow $t_{\text{BH}} \simeq 11 M + \tau \ln(10^{-2}/\varepsilon)$.

M	R_*	τ	time to a horizon		
			$\varepsilon = 10^{-2}$	10^{-3}	noise
$30 M_\odot$	172 km	0.87 ms	1.6 ms	3.7 ms	9.0 ms
$10^3 M_\odot$	5.7×10^3 km	29 ms	54 ms	0.12 s	0.30 s
$10^6 M_\odot$	5.7×10^6 km	29 s	54 s	2.1 min	5.0 min
$10^9 M_\odot$	38 AU	8.1 h	15 h	1.4 d	3.5 d

peaks at 0.031 over $t = 70$ –80 and falls to 0.0093 over $t = 90$ –100, a doubling time lengthening from 22 to 74 units, and the kicked arm does the same thing earlier and harder, peaking at 0.036 over $t = 30$ –40 and reaching 0.0023 over $t = 90$ –100 (doubling time $19 \rightarrow 298$). The two branches are not mirror images: the collapse runs away to a horizon, the inflation coasts. Mass sets the stability margin from below: the lone $m = 0.5$ throat tracks its exact solution only to $t \approx 17$ at the 0.1% level (the $m = 1$ twin holds to $t \approx 35$) and dies at its origin at $t = 26.3$ with no horizon—the lighter throat is the less stable one.

E. Lifetime and physical scale

At fixed m/a the solution is scale-free, so every time above is a multiple of the throat’s own light-crossing time: the measured e-fold is $\tau\alpha_{\text{th}} = 0.78$ –0.87 R_* in proper time at the minimal surface, one crossing of the throat to within the resolution spread. The lifetime is then a logarithm of the seed. The two declared kicks fix it directly—a horizon at $11 M$ for $\varepsilon = 10^{-2}$ and $25 M$ for 10^{-3} , a decade costing $14 M \simeq \tau \ln 10$ —and the unkicked throat’s $61 M$ places the level-3 truncation seed at $\varepsilon \simeq 2 \times 10^{-6}$ on the same line. Table II carries those three measurements into physical units.

Nothing in the range survives. A stellar-mass drainhole is gone in milliseconds, a supermassive one in minutes, and only at $10^9 M_\odot$ does the lifetime reach days—against a Hubble time in every case. The logarithm is what makes this robust rather than a statement about our grids: ten further decades of quiet initial data extend a $30 M_\odot$ throat from 1.6 to 22 ms. This is the single-throat form of the budget that forbids an inspiral in Sec. XI A, and it is why a foam-born pair must collide as it is born or not at all.

V. TWO THROATS AT REST

A. The sign rule

Two identical throats released from rest push apart; reversing one throat’s scalar field turns the push into a pull (Fig. 4(a,b)). That one sign is the only gravity-driven route to a merger in this model, and every merging binary in this paper carries it. The exclusion is complete rather than marginal: both forces fall as $1/d^2$, so their ratio $(a^2 + m^2)/m^2 > 1$ is separation-independent and holds for *every* massive drainhole—no gap, no mass, no approach speed lets a like-signed pair fall together under its own gravity, and $\sigma = +1$ is simply not a merger channel. Nor is the flip a tuning put in to get one: $\phi \rightarrow -\phi$ leaves T_{ab} of Eq. (1), and with it the whole one-body geometry, untouched, so the $\sigma = -1$ partner is the same wormhole carrying the opposite scalar charge. The merger channel costs nothing at the level of the constituents. The measured pull-to-push ratio at $a = 2$ is 1.518 ± 0.021 over 15 slices against the predicted $3/2$, and the force follows the throat *width*—the scalar charge—not the mass, which is equal in all arms.

B. The force law

Four like-signed pairs, byte-identical but for the gap, give displacements at a common $t = 11.5$ of $0.4696/0.3699/0.2980/0.2438$ for $d = 12/14/16/18$. Pure $1/d^2$ is excluded— $F d^2$ rises monotonically $67.6 \rightarrow 79.0$ across the ladder, a 15.4% spread—but $F \propto (d + \delta)^{-2}$ with $\delta \simeq 3.5$ fits every rung pair, and it was a blind prediction: δ fitted on $d = 12$ –16 alone predicts 0.243 at $d = 18$ where inverse-square predicts 0.209, and the measurement is 0.2438. Since δ is comparable to the measured mouth radius of the placed pair (4.29 at $a = 2$, $d = 12$; $R_* = 3.89$ isolated, Sec. V C), the force is inverse-square in an *effective* separation exceeding the coordinate one by about a throat radius—the deviation lives in the distance label, not in the law. Across widths the push grows at every rung ($0.147/0.283/0.416/0.615$ for $a = 1/1.5/2/3$) with a measured exponent ≈ 1.36 and falling; the point-charge a^2 scaling is not confirmed.

C. Placement systematics

The superposition itself is on the ruler: freshly placed pairs read +1.8% ($d = 48$) to +20.5% ($d = 6$) wider than the isolated throat, falling as $d^{-1.16}$ (Fig. 4(c,d)), so every same- d comparison in this paper is made against that curve rather than against the lone throat. Read that way, the released head-on pair’s mouths are genuinely squeezed, -0.6% at $t = 7$ growing to -4.0% by $t = 13$ —an interaction effect, not placement, since the lone throat is exact until $t = 35$.

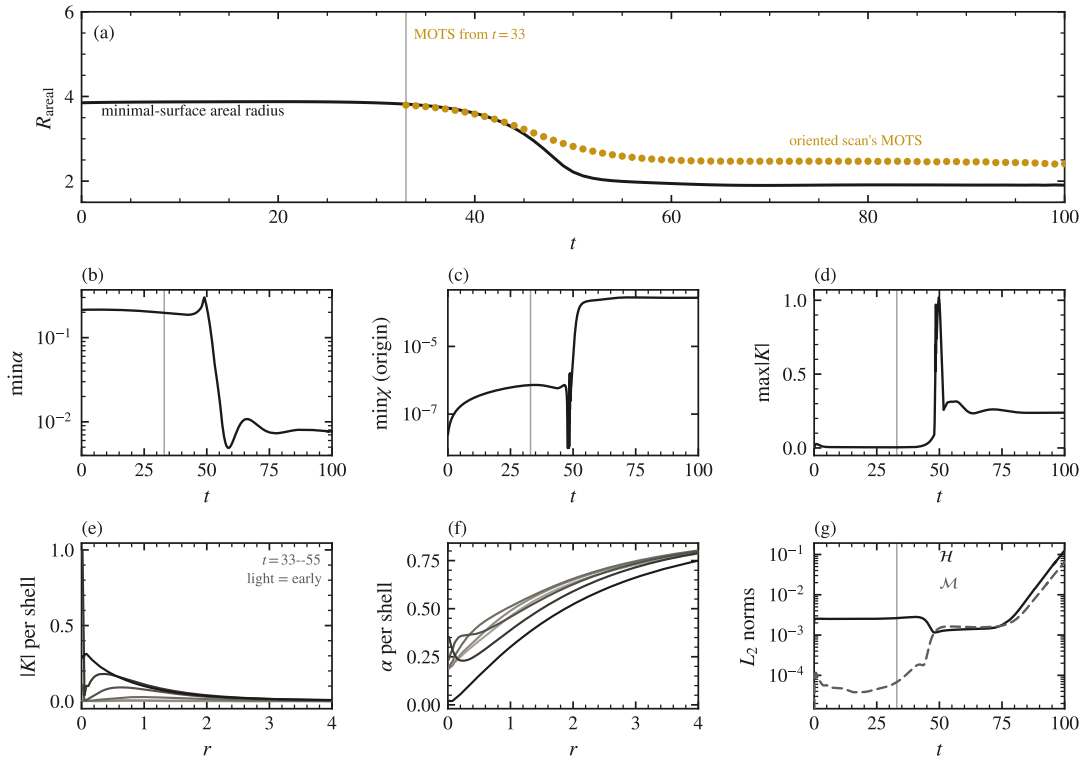


FIG. 2. The pure quadrupole’s collapse, fully instrumented (‘no kick’ arm: $\varepsilon_2 = 0.01$, radial kick zero). (a) Minimal-surface areal radius (ink) and the oriented scan’s MOTS (gold) from $t = 33$: R falls $3.80 \rightarrow 2.41$ and $M_{\text{MS}} 1.90 \rightarrow 1.24$ —phantom infall removes mass. (b)–(d) Global $\min \alpha$, $\min \chi$ (the origin monitor, the far sheet’s compactified infinity) and $\max |K|$; the violent phase is $t = 45$ – 55 . (e)–(f) The radial profile’s 128 shells over $t = 33$ – 55 (light = early): $|K|$ grows inward while α pinches to zero at the centre. (g) Constraint norms; the late rise from $t \gtrsim 75$ is this family’s known sphere-local floor growth, shared with the unkicked control.

VI. THE HEAD-ON MERGER

A. Contact and the common horizon

Released from rest at $d = 8$ with opposite scalar signs, the pair closes to contact at $t \approx 20$ while both mouths are still open wormholes, and a common marginally outer-trapped surface encloses them from $t = 22$: the offline corrected scan finds it at $r = 3.17$, areal radius $R = 5.56$, Misner–Sharp mass 2.99, growing to $R = 5.71$ by $t = 24$ (Fig. 5). The per-mouth scans never find a surface at either mouth at any time—the horizon is born common or not at all. The formation is robust where the wall is not. The frozen-interior ensemble carrying $\varepsilon = \pm 10^{-2}$ on both throats—the amplitude that decided a *lone* throat’s fate—finds the same surface at $t = 22.0$, with r_{mots} and M_{MS} within half a percent of the unkicked arm, and the $-\varepsilon$ member tracks that horizon to $t = 100$ within $\sim 3\%$ in radius and $\sim 1\%$ in mass; the same kick moves the level-3 death by more than a unit ($+\varepsilon$ dies at $t = 25.76$ against the scout’s 26.91). The seed sets the throat’s fate and the grid’s lifetime; it does not set the pair’s.

B. Through the wall to a black hole

Five units after the horizon forms, the level-3 scout dies at $t = 26.91$ with the classic anatomy— χ floored at the midpoint, $\max |K|$ running away to 18, a one-step overflow—but the failure is the grid’s, not the space-time’s: at $t = 27.05$, where the scout died, the level-5 arm reads a smooth, monotone midpoint (lapse 1.6×10^{-2} , $\chi = 1.2 \times 10^{-6}$, $\max |K| = 0.11$) and runs to $t = 100$ with the constraint norms *falling* the whole way. The remnant is a trapped ball with no throat left, losing mass to the phantom field it swallows, and it settles: the tracked M_{MS} fits $M_\infty + b e^{-(t-36)/\tau}$ with $M_\infty = 2.160 \pm 0.006$ and $\tau = 19.4 \pm 0.8$, a factor 4.8 better than a straight line, while the horizon radius stays linear over the same record with no asymptote resolvable (Fig. 5).

The arm quoted here carries level 5 from $t = 0$: no restart, no mesh seam, no interior fill anywhere on the record. It supersedes the level-3 $\rightarrow$ 5 chain used in earlier drafts, and reproduces it—the two arms’ (2, 0) waveforms agree to 2.9–5.4% of peak over the burst window $t = 23$ – 70 at $R = 10/14/18$, and their radiated energies to 3–8%; only the post-burst tail, where both amplitudes are below 4×10^{-4} and both arms are noisy, diverges (up to

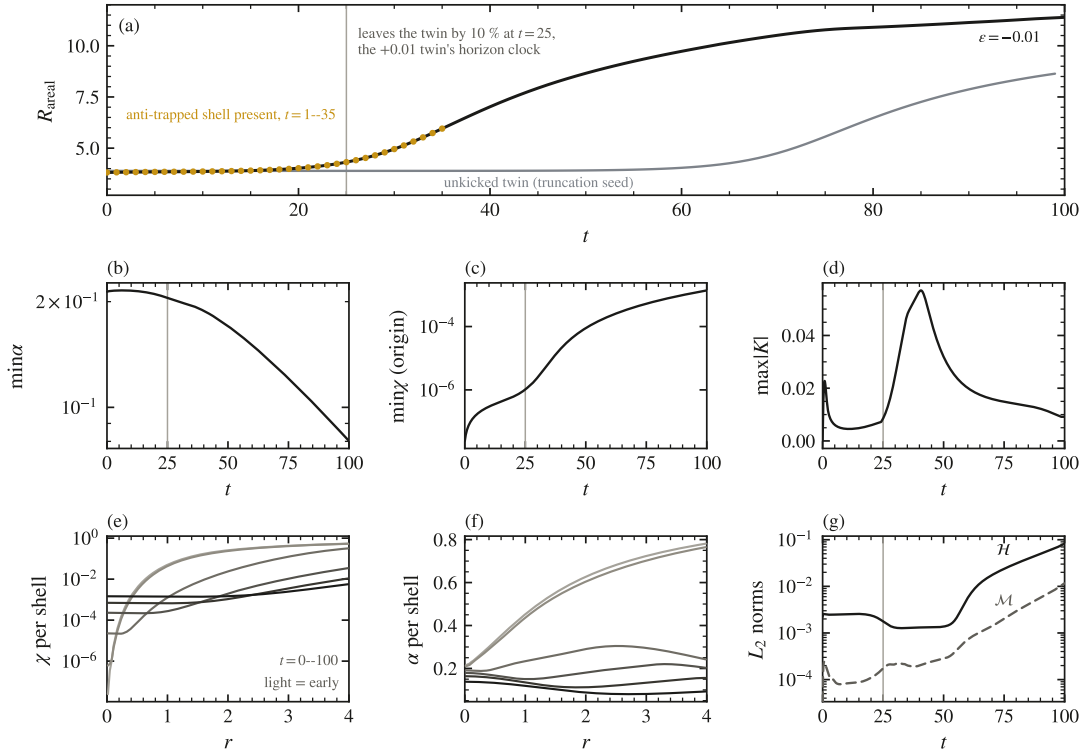


FIG. 3. The inflation branch, fully instrumented ($\varepsilon = -10^{-2}$, level 4), the mirror of Fig. 2. (a) Minimal-surface areal radius (ink) against the unkicked level-4 twin (grey), whose own truncation seed inflates it: $3.81 \rightarrow 11.39$ ($\times 3.0$) against $\times 2.2$, the two parting by 10% at $t = 25$ —the time by which the $+10^{-2}$ twin on this level has already formed its horizon. Gold: the scan rows carrying an anti-trapped shell, $\theta_+ > 0$ and $\theta_- > 0$, the marginally trapped surface’s mirror, held from $t = 1$ to 35. (b)–(d) Gauge and matter at the origin: $\min \alpha$ falls $0.214 \rightarrow 0.080$, $\max |K|$ peaks at 0.057 at $t = 40.6$ and then decays, and $\min \chi$ rises $2.4 \times 10^{-8} \rightarrow 1.4 \times 10^{-3}$ —the opposite of the collapse branch in every one. (e),(f) Shell profiles to $r = 4$, light = early: the compactified inner sheet’s χ -trough marches outward, $r = 0.016 \rightarrow 2.9$. That march is what the flickering $\theta_+ = 0$ rows at $R \approx 61$ after $t = 85$ are—the far universe’s image crossing the scan window—and not a horizon. (g) Constraints hold to $t \simeq 50$ and then grow, $60\times$ in $\mathcal{H}$ by $t = 100$; the record is quoted only where the mode is exponential.

62% at $R = 18$). The seam was therefore never carrying the result. The exterior is grid- and device-independent on the same evidence: the down-step arm agrees on the horizon mass to 1.3% and on the wave to 0.05–0.33% of peak over $t = 45\text{--}100$, and the fill twins agree with the no-fill arms at the 0.01%-of-peak level.

C. The half-mass pair

Halving the throat mass does not avoid the horizon; it moves everything earlier. At $m = 0.5$, $d = 6$, the lapse falls from 0.44 to 0.036 over $t = 7.5\text{--}13$, the tracker merges the mouths at $t = 11.1$, the offline scan confirms a common MOTS at $t = 12$ (areal $R = 3.83$, $M_{\text{MS}} = 2.16$), and the run dies at $t = 14.0$. The death belongs to the pair: the lone $m = 0.5$ control is exact to 0.02% through $t = 14$ and only fails, with no horizon of its own, at $t = 26.3$.

VII. THE SPIRAL MERGER

A. Capture and scattering

Dialling the tangential momentum at $d = 12$ splits the outcomes three ways: $p = 0.12$ and 0.15 fuse ($p = 0.15$ reproduces $p = 0.12$ ’s endgame signature—a plateau at pit separation 0.816 against 0.815, then the dive with the core lapse rising); $p = 0.20$ and 0.25 are captured into bound orbits; $p = 0.35$ and 0.45 escape. The capture boundary therefore sits between $p = 0.25$ and 0.35 —50–70% of the circular value—and the fusing boundary between 0.15 and 0.20 . The $p = 0.45$ fly-by is its own result, and every number quoted for it in this paper comes from one arm: the $L = 128$, level-5 record that also supplies its waveform. On that arm the pair reaches closest approach 4.82 at $t = 40$ and is back out at 9.6 by $t = 100$, still separating—a scattering pass, not a bound one. (An earlier $L = 64$, level-3 arm read 3.95 at $t = 40$ and came back in; it is superseded, and its number should not be read beside the level-5 one.) After the pass

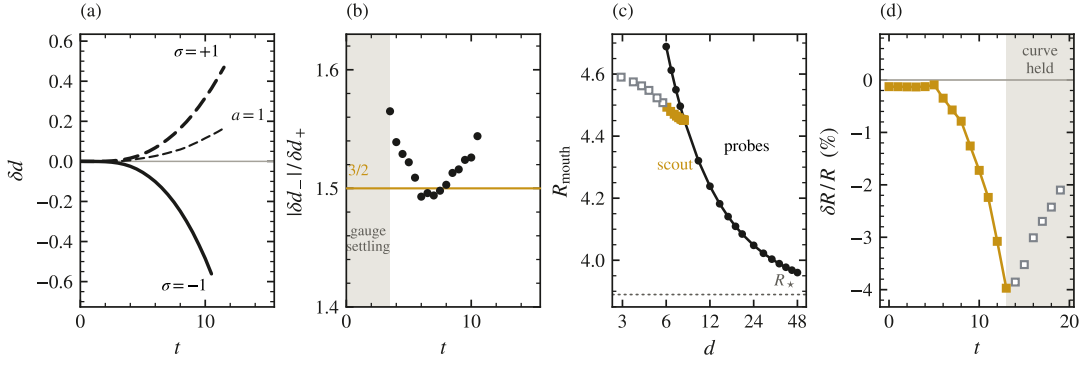


FIG. 4. Two throats at rest: the sign rule (a,b) and the ruler it is measured with (c,d). (a) Separation change of three $d = 12$ pairs one knob apart: the like pair ($\sigma = +1$, heavy dashed) pushes apart, flipping one field ($\sigma = -1$, solid) turns the push into a pull, and the narrow like pair ($a = 1$, light dashed) opens $2.8\times$ more slowly—the force follows the throat width, not the mass. Separations are inverse- χ -weighted centroids of the midplane χ pits, sub-cell where the tracker’s 0.03 grid quantum would swamp the early motion. (b) Pull over push: with $Q \equiv |F_\phi/F_{\text{grav}}| = 5$ (Sec. II C), the prediction is $(Q+1)/(Q-1) = 3/2$ (gold); measured 1.518 ± 0.021 . The gauge-settling systematic (grey band) is common to both arms and cancels in the ratio. (c) Per-mouth areal radius of freshly placed pairs at $t = 0$ (ink circles): wider than the isolated throat (R_* , dotted) by an excess falling as $d^{-1.16}$ —the neighbour’s field on the ruler. Gold squares: the head-on scout’s mouths (Sec. VI) at the separation reached; open grey squares (here and in d) lie closer than the closest probe, where no calibration backs the comparison. (d) The scout against the curve, $\delta R/R = R_{\text{scout}}/R_{\text{place}} - 1$: the -0.13% plateau to $t = 5$ calibrates the tracker’s grid snap, and the throats’ own squeeze then grows to -4.0% by $t = 13$. In the shaded band the curve is held at its last point, so the magnitude is a lower bound; a boost can only raise a coordinate-sphere reading, so a reading below the curve is a real squeeze.

no horizon ever forms—the corrected scan reads zero surfaces on the whole record—and both mouths swell $\times 7.8$ in areal radius, that expansion, not any wall, driving the late constraint growth ($\text{corr}(\log R, \log \|\mathcal{H}\|) = 0.92$). Past $t \approx 60$ the mouths are large enough that “separation” is no longer a clean observable at all: the areal radius reaches 33 by $t = 97$, so the tracked pit centres, not the throats, are what the number describes.

The swelling is the constituents’ own unstable mode with the companion driving it, and the honest measure of the driving is the mode’s e-fold time, not the growth rate of the radius. Fitted on the growth *excess* $R/R_0 - 1$, which is the quantity that grows exponentially, the mode runs at $\tau = 5.9$ in isolation (Sec. IV B), 4.39 in this fly-by and 3.70 in the arm that merges (Sec. VII B): a companion shortens the throat’s clock by 25–37%. The much larger factors one reads off $d \ln R/dt$ itself—399 units early, 52 through the approach, 32 after closest passage—are mostly not tidal driving at all: a mode of *fixed* τ accelerates the radius automatically as its amplitude climbs, and only the residual 25–37% is the companion. It is the τ that the lifetime budget of Sec. XI A runs on.

B. Plunge and collapse

Every “spiral” in this campaign is a plunge, not an inspiral: with $M_{\text{ADM}} \simeq 1$ per throat, $d = 12$ is $d/M = 6$ —the innermost-stable-circular-orbit separation—and the throat tracks of the momentum scan complete only 0.16–0.23 of a revolution before closest approach ($p = 0.20$ –0.35). Whether a wider pair could inspiral instead

is answered, in the negative, by the lifetime budget of Sec. XI A. The production spiral ($p = 0.12$, $L = 128$) falls from $d = 11.9$ to a single merged pit and then collapses at the centre while the throat stays open (Fig. 7). The merger’s constraint transient at $t = 41.2$ is resolved, not resolution—the level-3 leg reproduces the level-5 norms to 3.5%/0.7% over the shared window—and everything after it is a clean gauge-and-curvature collapse: the lapse to 2.0×10^{-3} , χ to its floor at $t = 58.4$, $\max |K|$ up $25\times$ over $t = 50$ –58 alone (and $\times 700$, $0.0084 \rightarrow 6.06$, across the whole level-5 window) on a wall that lands on the throat. Unlike the head-on, the corrected scans find *no* marginally outer-trapped surface at any of $t = 55$ –60, at either resolution, while the throat’s areal radius shrinks only 5.7%: at its death this remnant is still a wormhole.

The throats are not passive while the orbit closes. A stage-1 re-run of the production arm carrying the per-mouth oriented scan—the same initial data, level 3 to $t = 50$ —measures each mouth growing from areal radius 4.2385 to 4.7567, $+12.2\%$, by $t = 28$, the last sample at which the two scan spheres are disjoint and each reading is still about one throat, while the separation closes $11.9 \rightarrow 3.6$ (Fig. 6). Fitted over $t = 8$ –25, the window used on the fly-by, the growth excess is exponential with e-fold time 3.70 against the fly-by’s 4.39: the two arms share their initial separation and their throats and differ in tangential momentum, one reaching contact and the other missing, yet their mouths inflate on the same clock to 19%—the instability belongs to the throat, not to the encounter, which is the premise the budget of Sec. XI A runs on. The two arms differ in refinement as well as in momentum (levels 3 and 5), so this is agreement across

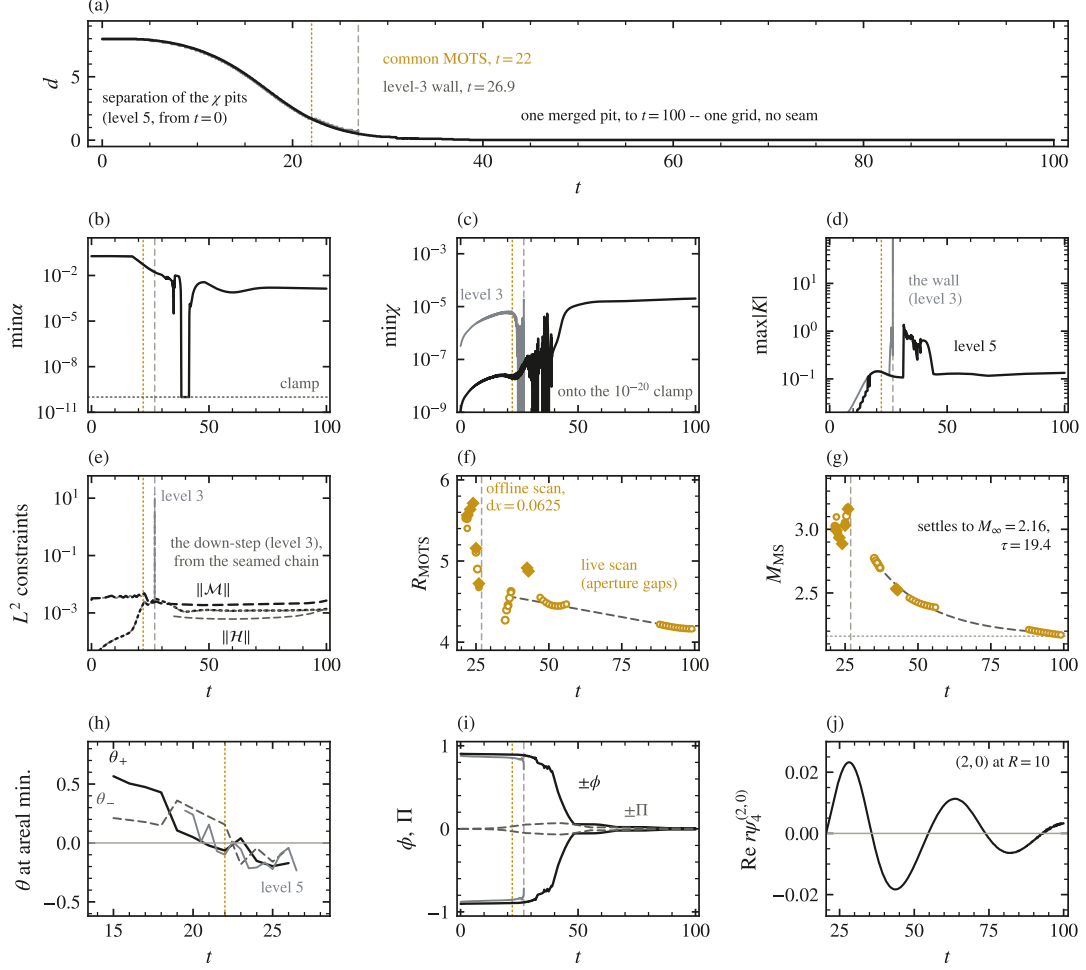


FIG. 5. The head-on merger makes a black hole ($d = 8$ from rest, opposite scalar signs), on the seamless arm: maximum level 5 from $t = 0$, no restart, no mesh seam, no interior device, $t = 0 \rightarrow 100$ with no NaN. It supersedes the level-3 $\rightarrow$ 5 chain of earlier drafts, which it reproduces (Sec. VIB). Every horizon point is the orientation-corrected scan; the level-3 scout is kept only to mark its wall. (a) Separation of the χ pits from $d = 7.94$ (the tracker's grid snap on a true 8): the common MOTS forms at $t = 22$ (gold rule in every time panel), the scout dies at the level-3 wall at $t = 26.91$ (dashed rule), and one merged pit runs on to $t = 100$. (b) The minimum lapse rides the 10^{-10} clamp over $t = 38.2-41.5$ and settles at 1.4×10^{-3} . (c) Minimum χ : excursions onto the 10^{-20} clamp leave the frame from $t = 30.6$; the arm returns and settles at 2.0×10^{-5} . (d) The wall is resolution: $\max |K|$ blows up to 18.1 at the level-3 death; level 5 peaks at 1.36 and ends at 0.13. (e) The scout's $\|\mathcal{H}\|$ explodes at the wall (13.6); the seamless arm's norms *fall* the whole way, $4.8 \times 10^{-3} \rightarrow 2.7 \times 10^{-3}$. The down-step arm (level 3, restarted from the seamed chain's $t = 35$ checkpoint) sits a constant ~ 0.2 dex below in $\|\mathcal{H}\|$ —a coarser grid's own truncation—while tracking the same horizon mass to 1.3%. (f) The horizon: filled diamonds, offline scans at $dx = 0.0625$ (formation at $t = 22$, $r = 3.17$, $R = 5.56$; growth to 5.71 by $t = 24$; then 4.72, and 4.92/4.87 at $t = 42.5/43$); open circles, the live scan on three arms. Its gaps are aperture, not horizon loss—the shells reach $0.5d + 2.3$, so once the pits merge the surface at $r \simeq 2.7-3.3$ lies outside while every shell within reports trapped; where two arms see it at once they agree ($t = 36$: $R = 4.456$ against 4.441). The dashed line is a straight-line fit to the tracked radius, -0.00658 per unit (r.m.s. 0.032); an exponential does not converge on this record, so no asymptotic radius is resolvable. (g) The Misner–Sharp mass falls, 2.99 at formation to 2.17 at $t = 99$ —phantom infall *removes* mass—but it does not fall without end. The dashed curve is $M_\infty + be^{-(t-36)/\tau}$ fitted on the tracked points: $M_\infty = 2.160 \pm 0.006$, $\tau = 19.4 \pm 0.8$ (r.m.s. 0.011, a factor 4.8 better than the straight line, dotted asymptote). The remnant is settling to a finite mass, and on the same clock as the scalar channel's shut-off, $\tau = 19-28$ (Fig. 14). (h) Expansions at the areal minimum: θ_+ crosses zero between $t = 20$ and 21, both negative by $t = 24$; the per-mouth scans never find an own MOTS—the horizon is born common. (i) The field's extrema fall $\pm 0.90 \rightarrow \pm 0.008$ with $|\Pi| \leq 0.07$: the supporting scalar is swallowed and radiated, leaving vacuum around a black hole. (j) The remnant's $(2,0)$ mode at $R = 10$, peak $r\psi_4 = +2.33 \times 10^{-2}$ at $t = 28.2$, rings with period $\sim 31-39$, $\times 0.6-0.8$ per half-swing, as the mass in (g) settles; the same number and the same convention as Sec. VIII and Fig. 9.

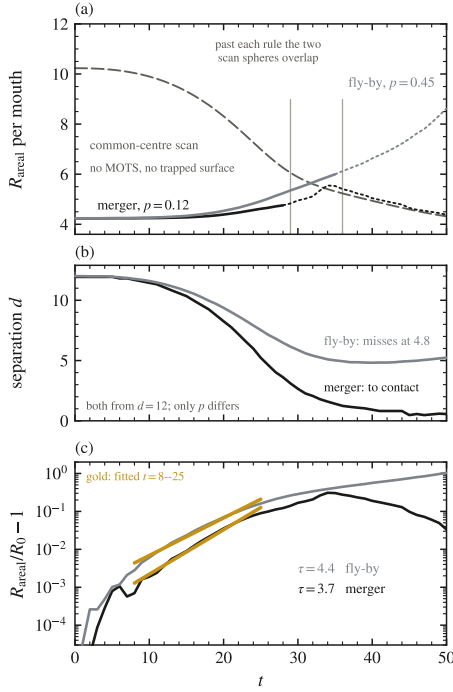


FIG. 6. The merging binary’s mouths, measured. Both arms start at $d = 12$ with the same throats and differ in tangential momentum, $p = 0.12$ against 0.45 ; they also differ in grid, the merger’s instrumented re-run carrying three refinement levels against the fly-by’s five, so the shared τ below is a result across two resolutions rather than a controlled one-knob comparison. (a) Areal radius of one mouth, its twin tracking it to 2×10^{-4} : solid while the two oriented scan spheres are disjoint, dotted once they overlap and the reading is a property of the pair; long dash, the common-centre scan enclosing both. No marginally trapped surface forms in either arm. (b) Separation: $p = 0.12$ reaches contact, $p = 0.45$ misses at 4.8 . (c) Growth excess $R_{\text{areal}}/R_0 - 1$ with exponential fits over $t = 8-25$ (gold), $\tau = 3.70$ for the merger against 4.39 for the fly-by—the arm that merges inflates on the clock of the arm that misses.

two grids rather than a single-knob comparison; what makes it usable is that the spiral’s own wave and scalar streams are level-independent to 0.005% over their shared window (Sec. IX A). Two mouths track each other to 2×10^{-4} in areal radius, so this is the radial mode and not an orbital asymmetry, and no marginally trapped or anti-trapped surface appears on any of the three scan centres through $t = 50$. The burst of Sec. VIII C is therefore radiated by throats already more than a tenth larger than the initial data’s by the time the pair has halved its separation.

C. The wall

No grid gets the spiral through its merger. Restarting the $p = 0.12$ run from the same $t = 50$ checkpoint at max_level 3–7, every arm dies on the same h_{11} NaN, and

the gains saturate: $+1.03$, $+2.51$, $+0.53$, $+0.06$ units per doubling (Fig. 8). Sixteen times the resolution buys 4.13 units and never reaches the merger’s end, and a halved time step buys $+0.36$ on its own level-5 twin—most of the level $5 \rightarrow 6$ spatial gain—so the top of the ladder is saturation evidence, not a converged limit. The wall is insensitive to the matter device (the damped family dies on the same clock), and to starting refined—the two arms that carry level 5 from $t = 0$ are $p = 0.20$ and $p = 0.25$, and both die at the level-3 wall ($t = 52.07$ and 52.79), so the gain per level belongs to the *restart* recipe and not to depth; the matching cell for $p = 0.12$ at $L = 128$, level 5 from birth, has not been run—and it moves only modestly when the whole problem is re-based: the production chain—base grid doubled, restarted three times earlier at $t = 36$ —buys 4.2 units past the entire ladder and still dies, at $t = 60.445$. The interior freeze then carries the exterior to $t = 100$; it is a numerical device, engaged only after the burst is causally clear of the extraction spheres (Sec. VIII C).

Why refinement rescues one merger and not the other has a one-line answer that every arm in Table I obeys: *resolution rescues a curvature wall only when a horizon censors it*. The head-on’s wall kills level 3 at $t = 26.91$, but a marginally outer-trapped surface has enclosed the site since $t = 22$, and level 5 walks through it and runs to $t = 100$ with the constraints falling (Sec. VI B). The single throat’s collapse wall is likewise behind a MOTS from $t = 33$, and that branch runs clean to $t = 100$ at level 3 alone. The spiral’s wall has no horizon anywhere near it—the corrected scan finds no trapped surface at any time, at any of levels 3 to 7, on any centre—and no grid survives it: sixteen times the resolution buys 4.13 units, and re-basing the whole problem buys 4.2 more and still dies. A last arm removes the remaining escape, that the wall might be the grid’s own truncation noise rather than the spacetime’s: run from the initial data to the wall at level 5 throughout—no restart, no seam, and therefore a seed two refinement levels smaller than the arm that inherits one from a level-3 checkpoint—the merger dies at $t = 59.94$ against that arm’s 60.45 , the same wall to 0.8% and, if anything, marginally sooner. Refinement applied earlier buys nothing, so the wall is not seeded by the grid. The pattern is not a numerical curiosity. A censored wall is a coordinate problem behind a surface whose exterior is causally insulated from it, so a finer grid has only to postpone a gauge failure the exterior never feels; a naked one is the exterior’s own geometry, and no cell size makes it integrable. On this reading the spiral’s death is a statement about the spacetime and the head-on’s is a statement about the grid—which is exactly how the two are reported here. One arm is untested against the rule rather than consistent with it: the half-mass head-on forms its MOTS at $t = 12$ and still dies at $t = 14.0$, but it was never re-run at level 5, so whether refinement would have carried it is unknown.

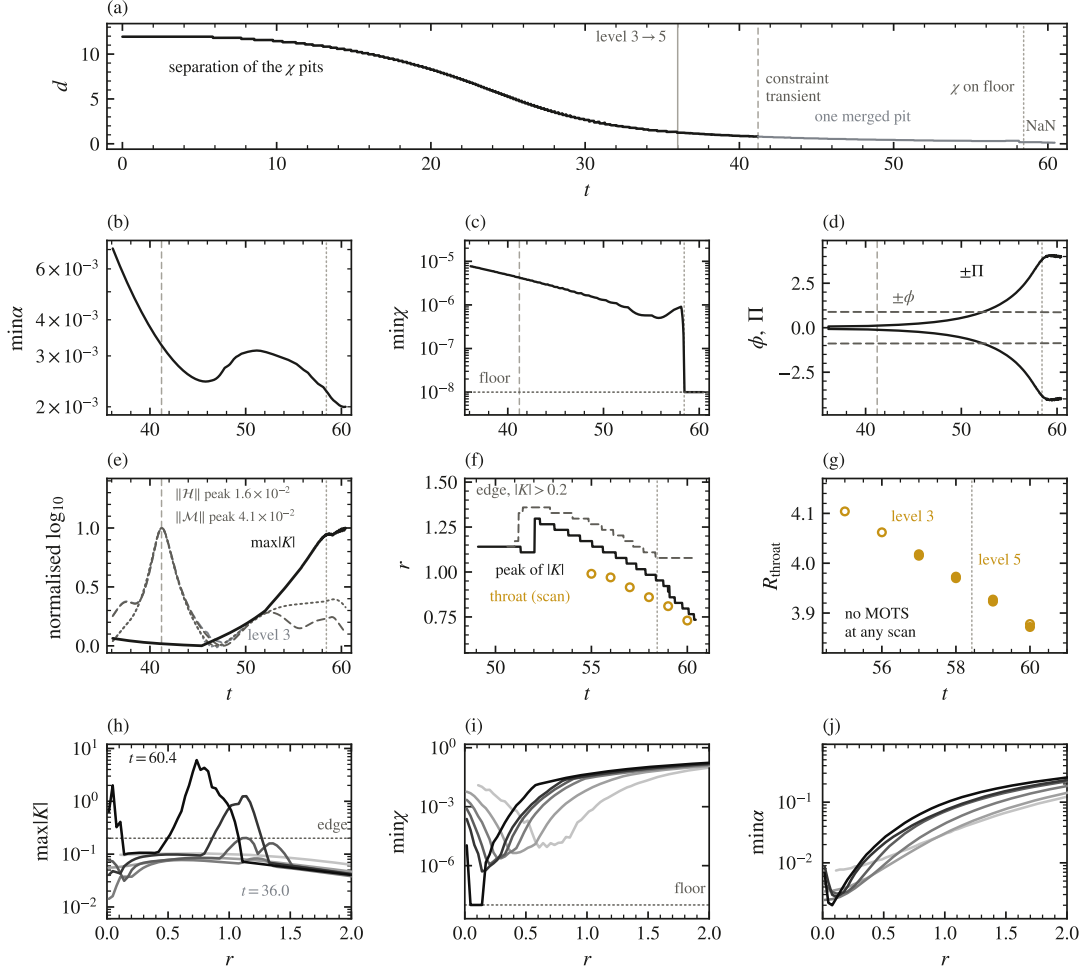


FIG. 7. The production spiral's collapse ($p = 0.12$, $d = 12$): level 5 from the restart at $t = 36$ to the death by NaN at $t = 60.44$, no panel involving the interior fill. Rules in every time panel: the handover at $t = 36$ (solid), the merger transient at $t = 41.2$ (dashed), the χ floor at $t = 58.43$ (dotted). (a) Separation of the two χ pits over the whole history—the level-3 leg carries the plunge from $d = 11.9$, after which the tracker follows one merged pit (pale). (b)–(d) $\min \alpha$ bottoms at 2.0×10^{-3} without reaching zero; $\min \chi$ decays exponentially onto its 10^{-8} floor at $r = 0.078$; the scalar holds ± 0.89 while Π saturates at ± 4.1 —the profile survives, the momentum diverges. (e) Growth compared, each curve $\log_{10}$ -scaled onto $[0, 1]$: the constraint norms peak at the merger and relax below where they started, while $\max |K|$ climbs only from $t = 49$, with $\text{corr}(\log \max |K|, \log \|\mathcal{H}\|) = -0.35$ after $t = 40$ —the death is not a constraint blow-up. Pale: the level-3 leg through the same transforms, median 3.5% ($\mathcal{H}$) and 0.7% ($\mathcal{M}$) over $t = 36$ –50. (f) Radius of the $|K|$ spike (steps, drawn from $t = 49.1$; earlier the profile is flat and an argmax is noise) and the $|K| > 0.2$ edge (dashed), widest at $r = 1.36$. Gold: the throat radius from the corrected scans—the spike rides just outside the throat and lands on it by the death. (g) Throat areal radius $4.10 \rightarrow 3.87$ over $t = 55$ –60, levels 3 and 5 agreeing to 0.2%, with *no* marginally outer-trapped surface at any scan. (The in-code θ_+ columns are omitted: their naive $+r$ orientation manufactures trapped verdicts near a throat.) (h)–(j) Radial anatomy at six equally spaced times, light ($t = 36.1$) to dark ($t = 60.4$): per-shell $\max |K|$ grows $0.08 \rightarrow 6.1$ on a spike ~ 0.25 wide migrating inward, while the χ and lapse pits stay at $r \lesssim 0.15$ —the gauge collapse and the curvature wall are different places.

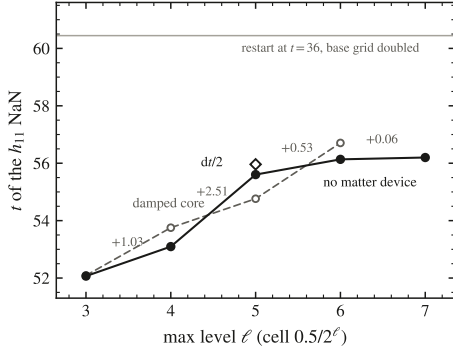


FIG. 8. The wall does not move: death time of the $p = 0.12$ spiral against refinement, every arm restarted from the same $t = 50$ checkpoint, every death an h_{11} NaN, no horizon at any rung. Filled: plain evolution, per-doubling gains marked. Open diamond: the level-5 arm at half the time step. Open circles: the core-damping family—a different treatment, dying on the same clock. Faint rule: the production chain (doubled base grid, restart at $t = 36$), 4.2 units past the whole ladder and still short of the merger’s end.

VIII. GRAVITATIONAL WAVES

Every channel radiates, and the loudest is the one that never makes a horizon. At the innermost sphere the peak $|r\Psi_4|$ of the dominant mode reads: fly-by 4.1×10^{-2} , spiral 2.9×10^{-2} , head-on 2.3×10^{-2} , vacuum twin 1.1×10^{-2} , collapsing throat 2.9×10^{-3} (Fig. 9)—the scattering pass outradiates every merger, by $1.4\times$ over the spiral and $3.8\times$ over the equal-parameter vacuum binary. Each record propagates at c within the output cadence and falls as $1/r$ across its spheres, and rescaled to a detector the ordering survives: at each record’s own resolved spectral peak the strain amplitude spectral density reads $1.2 \times 10^{-21} \text{ Hz}^{-1/2}$ for the fly-by (193 Hz), 1.1×10^{-21} for the head-on and 9.3×10^{-22} for the spiral (both 406 Hz), 3.0×10^{-22} for the vacuum twin (450 Hz) and 1.5×10^{-22} for the lone throat (286 Hz) (Fig. 10). These are quoted at a stated frequency and not as maxima of the curve, because these bursts have no resolved strain maximum: after the $1/f^4$ weight every one of them still rises toward low frequency, so the maximum sits wherever the record’s own low-frequency corner stops it—a statement about record length rather than about the source.

The energy budget makes the same point without a detector: in the dominant multipole the fly-by radiates $7.4 \times 10^{-2} M$ and the spiral $2.2 \times 10^{-2} M$, against $2.6 \times 10^{-3} M$ for this campaign’s vacuum merger control— $29\times$ and $8.5\times$ more. That control matches the *spiral* in every parameter, $d = 12$, $p = 0.12$, equal masses, and is therefore the wrong denominator for the fly-by. The right one was run: a vacuum binary with the fly-by’s own d and $p = 0.45$ and the same bare masses. In vacuum that momentum is unbound—the Newtonian circular value at $d = 12$ is 0.204 and the parabolic one 0.289—so where the drainhole pair fell from 12 to 4.8 on a pull six times

gravity, the black holes start at periapsis and coast apart, sweeping 96° while their separation opens $12 \rightarrow 21$. They radiate $1.05 \times 10^{-3} M$, a factor 2.3 *below* the merger control on the sphere where both are in the wave zone. The fly-by therefore outradiates a vacuum binary of its own parameters by $\simeq 70\times$, not 29; the published 29 is the value against a vacuum *merger* and is retained above because it is the comparison the literature’s quasi-circular numbers invite. The fly-by also exceeds the $4.84 \times 10^{-2} M$ a quasi-circular vacuum merger of the same mass radiates from infinity. That bound brackets vacuum binaries because it is a statement about a positive-energy system with a finite reservoir; a ghost field is unbounded below, so no such bracket applies here and the exceedance is a signature of the energy condition being violated rather than a symptom of a contaminated measurement—the sphere spread, quoted below, is the instrument for the latter.

A. The collapsing throat

A lone throat radiates only if its collapse is not spherical. The spherically kicked arm emits nothing above the $\ell = 2$ numerical floor—by symmetry—while the same kick dressed with an $\varepsilon_2 = 0.05$ quadrupole produces a burst of peak $|r\Psi_4| = 2.9 \times 10^{-3}$, $35\times$ the no-quadrupole control, carrying $E_{\text{rad}} = 3.2 \times 10^{-5} M$. The burst arrives in radius order at $0.8\text{--}1.0c$, carries the same $r\Psi_4$ to four spheres within 7.2%, and scales by 4.85 for a $5\times$ kick. Two further arms close the case (Fig. 11): halving the seed halves the wave (ratios $1.99/1.97/1.87$ at $R = 10/14/18$), making the decade $\varepsilon_2 = 0.005\text{--}0.05$ linear to 1–16%; and a *pure* quadrupole—the same $\varepsilon_2 = 0.01$ with the radial kick removed—radiates the same wave to 2–3% while its throat still collapses (MOTS at $t = 33$, on the refinement level whose own truncation seed *inflates* the undeformed throat). The deformation, not the kick, is both the source of the wave and the selector of the fate. What is left rings at period $20.6 M$, 21% short of the Schwarzschild fundamental for the horizon mass; the decay time is not measurable on this record.

B. The head-on ringdown

The head-on’s signal is a bell with no chirp: the $(2,0)$ breathing mode at $R = 10$ swings through $+0.23, -0.18, +0.11, -0.07$ at $t = 28.2, 43.6, 63.0, 81.7$ —a period of $\sim 31\text{--}38$, lengthening, with the amplitude falling $\times 0.6\text{--}0.8$ per half-swing—and the same bell arrives ~ 4 units later per 4 units of extraction radius ($0.94c$), outgoing across all three spheres. On the detector track it is the bell that slows from 486 to 342 Hz in Fig. 10(c).

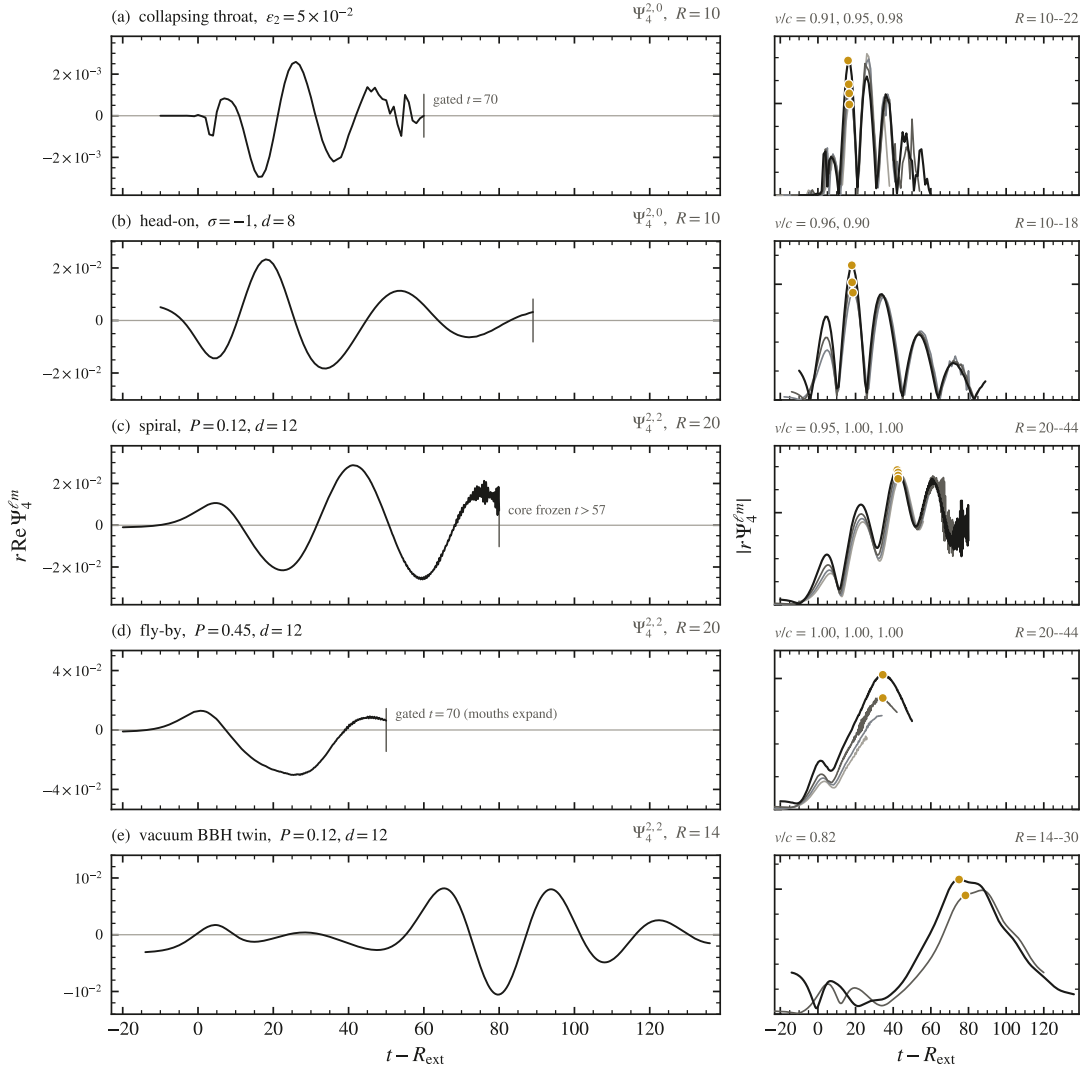


FIG. 9. The waveform gallery: every channel’s radiation, one row per scenario, one retarded clock. Left column: $r \text{Re} \Psi_4$ of the dominant mode at the innermost extraction sphere, each row on its own stated scale—the lone throat radiates an order of magnitude below the mergers, and the fly-by is the loudest channel of all. Half-height caps mark record ends: the collapsing throat’s stream is junk-gated at $t = 70$, the spiral is drawn to $t = 60$ —the end of its no-fill profiled leg, past which the record continues only behind a core frozen at $t = 57$ (Sec. VII C), so the panel stops where the interior is still evolved—and the fly-by—complete to $t = 100$, no horizon ever forming—is gated at $t = 70$, ahead of the $t = 76.1$ trough where the mouths’ expansion (areal radii growing $\times 7.8$) has grown to equal the decaying burst at the $R = 20$ sphere. Right column: the envelope $|\Psi_4^m|$ at every extraction sphere on its own retarded time (ink to faint, inner to outer): a signal travelling at c and falling as $1/r$ folds onto one curve. Gold dots mark the correlated wavefront, and the quoted v/c is the lag of the whole complex waveform between neighbouring spheres over their common retarded window—peak-to-peak timing fails on flat-topped envelopes (the vacuum twin’s merger plateau is ~ 15 units wide at both spheres, and its peak times alone would read $v = 0.57c$ where the waveform lag reads $0.82c$). Measured speeds: throat $0.91/0.95/0.98$, head-on $0.96/0.90$, spiral $0.95/1.00/1.00$, fly-by $1.00/1.00/1.00$, vacuum twin 0.82 . Every channel propagates at c within the output cadence: radiation, not a gauge or constraint mode.

C. The spiral’s carried burst

The spiral’s $(2, 2)$ burst peaks at $|r \Psi_4| = 2.9 \times 10^{-2}$ and folds onto one curve across $R = 20$ – 44 at $v/c = 0.95/1.00/1.00$. The burst is fill-free by causality: the interior freeze that carries the late record is armed at $t = 53$ with its skin at $r \geq 1.8$, so the earliest it can influence any extraction sphere is after the burst has passed it—

and the head-on’s fill twins bound the device’s imprint at the 0.01%-of-peak level. The production arm now carries its own certificate (Fig. 12): a twin restarted from the same $t = 57$ checkpoint with only the fill window moved, $1.40/1.90 \rightarrow 1.25/1.75$, shifts the $(2, 2)$ waveform by at most 0.385% of its peak at $R = 20$ and 0.013% at $R = 44$ over the whole shared record $t = 57$ – 100 (overlap > 0.999999), with the residual fronts arriving on the fill’s

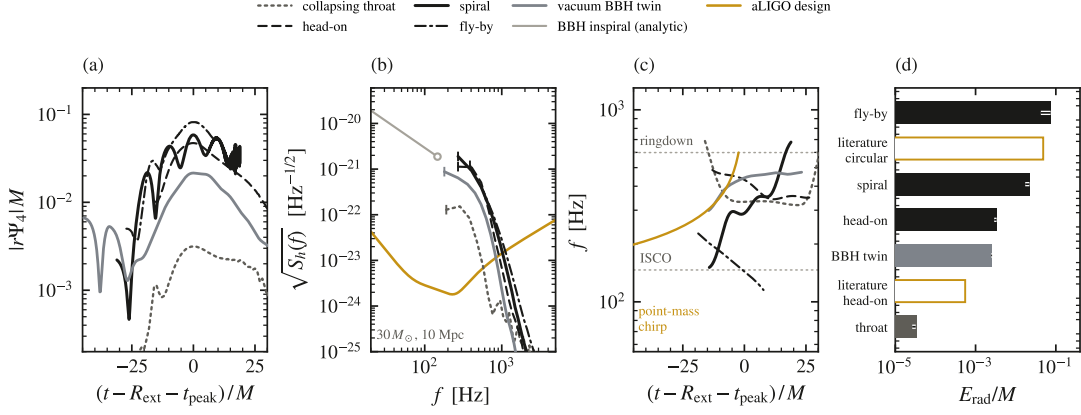


FIG. 10. Every source on a detector at one shared calibration: total mass $M = 30 M_\odot$, distance $D = 10$ Mpc, each record first reduced to units of its own total ADM mass; strain scales as M/D , frequency as $1/M$. (a) Envelopes $|r\Psi_4|/M$ at the innermost sphere on a common merger clock ($\tau = 0$ at each peak): fly-by loudest (8.2×10^{-2}), then spiral, head-on, vacuum twin, lone throat (3.1×10^{-3}). (b) Strain amplitude spectral densities over the aLIGO design floor (gold), each curve drawn from its own low-frequency corner (one cycle per record, ticked), below which a record of length T cannot tell a wave from a drift. Read them as falling power laws (slopes -3.3 to -5.9) with no resolved maximum: any “peak” taken from this panel is the corner $1/T$; the values at the resolved Ψ_4 peaks are quoted in the text. Faint line: the Newtonian inspiral of the same binary, $|\dot{h}| \propto f^{-7/6}$ to its ISCO at 147 Hz—the ramp an astrophysical binary rides into the band where the simulated signal begins. (c) Instantaneous frequency over each record’s body against the point-mass chirp (gold), the ISCO (147 Hz) and the equal-mass Kerr ringdown (599 Hz): the vacuum twin climbs the chirp and peaks at $fM = 0.065$, the textbook merger value; no wormhole channel does (tracks in Sec. XC). (d) Energy radiated in the dominant multipole, ranked; the tick on each bar is the spread over extraction spheres, the error bar that catches near-zone content posing as a wave (fly-by 43%, the widest: the expansion of its open mouths is near-zone content the outer spheres see less of). Open gold bars: the published equal-mass vacuum results from rest at infinity—head-on infall 5.5×10^{-4} , quasi-circular merger 4.84×10^{-2} —which bracket any *vacuum* binary and need not bracket a ghost one; the spiral and fly-by outradiate the vacuum merger control by $8.5\times$ and $29\times$. That control carries the spiral’s momentum; against the momentum-matched vacuum pass of Sec. VIII the fly-by’s factor is $\simeq 70$. Energies are band integrals of $|\int r\Psi_4 dt|^2$ over a common retarded window, doubled for the $\pm m$ pair; gravitational channel only—the scalar’s own energy is invisible to Ψ_4 and is measured separately in Sec. VIIF, where it is larger than every bar here and of the opposite sign.

own causal clock $t = 57 + (R - 1.9)$. The fill is a numerical device, not a source, and the carried burst survives having its container reshaped. What the burst is *not* is a black-hole birth cry: the corrected scans find no horizon anywhere on the record (Sec. VIIB), so this is the wave of a merger that stalls as a wormhole.

D. The fly-by burst

The loudest signal in the campaign comes from the encounter that makes nothing: the $p = 0.45$ pass emits a single $(2, 2)$ arch of peak $|r\Psi_4| = 4.1 \times 10^{-2}$ with no ringdown—there is no remnant to ring—propagating at $v/c = 1.00$ across every sphere pair. The record is complete to $t = 100$ with no NaN and no horizon at any time; it is gated at $t = 70$. The mouths’ post-encounter expansion (areal radii $4.2 \rightarrow 33$ by $t = 97$, and $\text{corr}(\log R, \log \|\mathcal{H}\|) = 0.92$ —the swelling, not a wall, drives the late constraint growth) contaminates the innermost sphere, whose $|r\Psi_4|$ stops decaying at $t = 76.1$ and ends at twice the burst peak by $t = 100$. That turning point is where the contaminant has grown to *equal* the decaying burst, not where it arrives, so the gate is set

ahead of it, 15.5 units past the $R = 20$ burst peak. The outer spheres are not a control here: in retarded time all four peak together at $t - R \simeq 34.5$, as radiation must.

E. The vacuum twin

The vacuum control—same d , p and masses, no scalar—does what black holes do: merges, climbs the point-mass chirp $299 \rightarrow 473$ Hz and peels off toward the equal-mass Kerr mode, with its Ψ_4 peak at the textbook $fM = 0.066$. Against it the wormhole channels are loud: the fly-by’s peak $|r\Psi_4|$ is $3.8\times$ the twin’s and the spiral’s $2.7\times$. The twin also calibrates the wavefront method itself: its merger envelope is a ~ 15 -unit plateau at both spheres, where peak-to-peak timing reads a meaningless $0.57c$ and the full-waveform lag reads $0.82c$ —which is why every speed in this chapter is a correlated lag, never a peak time.

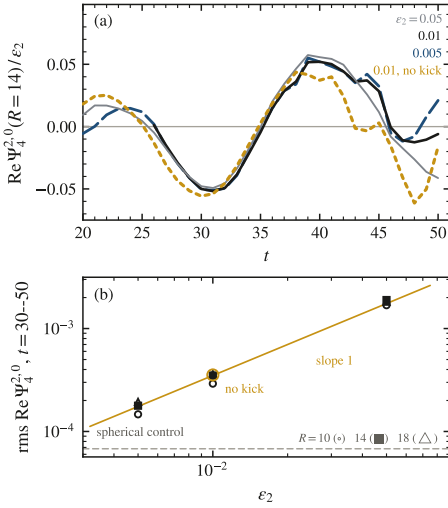


FIG. 11. One throat’s wave is linear in its deformation. (a) $\text{Re } \Psi_4^{2,0}$ at $R = 14$, each arm divided by its own seed: $\varepsilon_2 = 0.005, 0.01, 0.05$ and the kick-free 0.01 arm collapse onto one curve. (b) rms amplitude over $t = 30\text{--}50$ against ε_2 at three spheres, with the slope-one line through the 0.01 arm: linear to 1–16% across a decade. The open circle is the pure quadrupole—no radial kick—matching the kicked arm to 2–3%; the dashed rule is the spherical control’s floor.

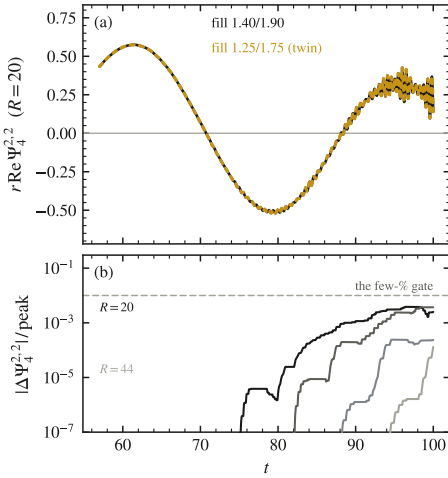


FIG. 12. Moving the fill window does not move the wave. (a) $r \text{Re } \Psi_4^{2,2}$ at $R = 20$ for the freeze arm (fill 1.40/1.90, ink) and its twin (1.25/1.75, gold dots), same $t = 57$ checkpoint, same binary: one bicolour curve. (b) The per-sphere envelope of the difference against the few-percent gate: at most 0.385% of peak at $R = 20$ and 0.013% at $R = 44$, each front arriving on the fill’s causal clock $t = 57 + (R - 1.9)$. The figure certifies the fill’s inertness, not a remnant ringdown.

F. The scalar channel, and its sign

Every energy above is a Ψ_4 energy, and Ψ_4 is spin-2: it cannot see a spin-0 wave. That blindness is not a detail for this matter model, because the throats are held

open by a field that carries charge, and a charge that is shaken radiates. Two opposite scalar charges—the sign rule of Sec. V makes every merging pair opposite—have a dipole moment, where a vacuum binary’s leading gravitational moment is the quadrupole. The scalar stream of Sec. III C rides the same extraction spheres as Ψ_4 on the spiral and fly-by arms, so the two channels can be compared sphere for sphere with no cross-pipeline normalisation in between (Fig. 13).

It is the dipole, and it is large. At $R = 30$, over the window in which both spheres still lie outside the inflating mouths, the scalar sector’s outgoing power $\sum_{\ell m} |\dot{A}_{\ell m}|^2$ is carried by $\ell = 1$ to within one part in 10^7 —the monopole and the quadrupole together are 10^{-7} of it—and the time-integrated kinematic flux reaches 2.3 times the gravitational-wave energy through the same sphere for the fly-by and 2.4 times for the spiral. The two arms agree although one merges and one misses; the two estimators of the scalar energy, the kinematic flux and the wave-zone mode sum, agree to a factor 1.7; and for the fly-by the inner sphere returns 3.0 against the outer sphere’s 2.3, a 30% spread narrower than the 43% its own Ψ_4 energy carries. The spiral’s inner sphere is not quoted: at $R = 14$ it sits only two units outside an orbit of initial diameter 12, and there both channels are near-zone—its Ψ_4 energy is $5\times$ the outer sphere’s and its scalar integral has not yet changed sign. The dominant radiation of a drainhole encounter is therefore not gravitational.

Its sign is the result. The stream is signed so that a *canonical* outgoing wave gives a positive flux; gravity here couples to minus that field’s stress tensor (Eq. (1)), so the physical energy flux is the negative of it, and what these arms radiate in $\ell = 1$ is negative energy. A bound system that emits negative energy gains: the dominant channel pumps the orbit where gravitational radiation damps it. This is the same bookkeeping, read at infinity, that the head-on remnant shows at its horizon, where swallowing the phantom field *removes* mass ($M_{\text{MS}} 3.0 \rightarrow 2.2$, Sec. VIA), and that the single-throat remnant shows in reverse when its horizon re-expands (Sec. IV D).

Three limits are declared with it. The flux is a coordinate flux—no lapse or shift factors are applied, which is justified at $R = 30$ where α and χ are within a percent of unity, and not elsewhere. Both spheres are inside the near zone for part of every record, which is why the number is quoted with its sphere spread and not to three digits. And the balance itself, $\dot{M}_{\text{ADM}} = -F_{\text{GW}} - F_\phi$, is *not* closed here: this campaign carries no surface-integral ADM mass, so what is established is the size, the multipole and the sign of the missing term, not that the books balance (Sec. XIB). The single-throat arms ran before the scalar stream existed and their plotfiles are not retained, so that channel has no number here; the head-on has since been re-run with the stream, and is the subject of the next section.

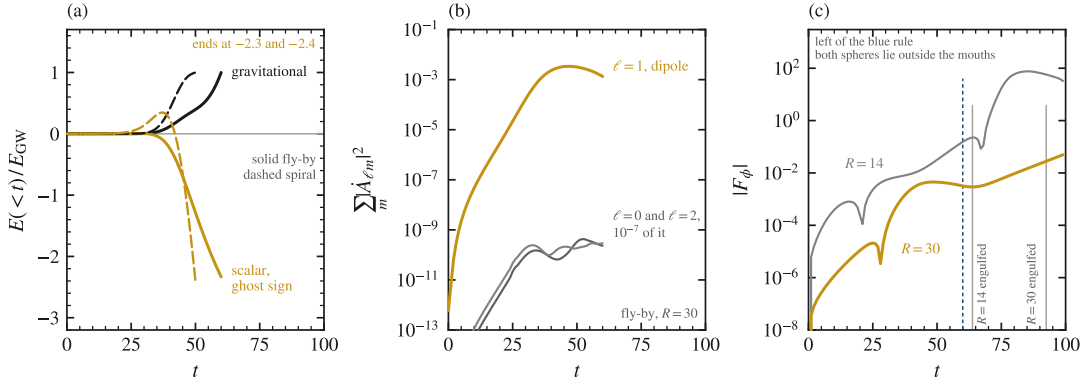


FIG. 13. The second radiation channel, on the same extraction spheres as Ψ_4 . (a) Energy through $R = 30$ against time, each arm in units of its own gravitational-wave energy at the rule: the gravitational curve rises to 1 by construction, the scalar curve falls to -2.3 (fly-by, solid) and -2.4 (spiral, dashed). The scalar energy is negative because gravity couples to minus this field’s stress tensor, so an outgoing scalar wave carries negative energy and the bound system gains. (b) Which multipole carries it: $\ell = 1$ against $\ell = 0$ and $\ell = 2$ for the fly-by, the dipole a factor 10^7 above both. Two opposite scalar charges radiate at dipole order; a vacuum binary has no such channel. (c) Where the instrument stops: $|F_\phi|$ at both spheres against the times at which the inflating mouths’ areal radius passes each sphere’s own radius (grey rules). Left of the blue rule both spheres lie outside the mouths, and the energies in (a) are taken there; the same window read at $R = 14$ gives 3.0 in place of 2.3, which is this measurement’s error bar.

G. The horizon censors it

The head-on arm has since been re-run at level 5 from $t = 0$ with the scalar stream on, which puts the same instrument on the one encounter in this campaign that makes a horizon. It turns the channel off (Fig. 14). What is left of the mouths’ hair after the common MOTS closes over them at $t \simeq 22$ leaves as a single $\ell = 1, m = 0$ pulse—the dipole carries the sector to one part in 10^7 here too—cresting later at each sphere in the causal order of an outgoing wave, and then the source is gone: log-linear fits over $t = 30\text{--}95$ give e-fold times $\tau = 19/23/28$ at $R = 10/14/18$. That is the remnant’s own clock. The horizon mass settles with $\tau = 19.4 \pm 0.8$ over the same stretch (Sec. VIB), so the hair is shed, the source goes quiet and the mass stops changing on one timescale. The post-horizon energies keep the sign of Sec. VIIF, $E_\phi = -0.056/-0.071/-0.075$ across the three spheres.

Two arms without a horizon make the control. The fly-by, which never merges, has a dipole envelope still *growing* at the end of its record; the $p = 0.12$ spiral, which merges and still forms no MOTS on any scan, is likewise climbing when its grid dies at $t = 59.9$. Horizon or no horizon is the only variable that separates them from the head-on, so the decay is the horizon’s doing and not the encounter’s.

Two cautions travel with this. Only the post-horizon window is integrated on the head-on: before $t = 22$ the flux at $R = 10$ runs canonically *ingoing*, which is the two open mouths’ static hair superposing inside the near zone and not radiation, so no full-record head-on integral appears anywhere in this paper. And past its wall the spiral’s curve is the frozen-core arm—its exterior is certified against the free arm to 0.1% on this flux over their

overlap, but a frozen source region cannot be asked how its radiation decays, so it is drawn dashed and carries no claim.

IX. CONTROLS AND SYSTEMATICS

A. Numerical systematics

Each numerical device carries a twin that measures it. The χ clamp is not load-bearing: twins at 10^{-8} against 10^{-20} re-converge to four digits after touching the floor, and the level-4 pair is byte-identical. The interior fill is not load-bearing for anything outside: head-on fill and no-fill arms agree to 0.01% of the wave peak. The spiral’s interior freeze is certified the same way, and directly rather than by proxy: over the 3.4 units in which the frozen-core arm and the no-fill profiled arm both exist, $t = 57.0\text{--}60.4$, the two agree in the $(2, 0)$ and $(2, 2)$ modes to 0.013% of peak on the innermost sphere, to one part in 10^{13} on the next, and *bit for bit* on the outer two—so what the device does not reach is not inferred from a window shift but read off an overlap. Over that same window the spiral’s burst is also resolution-independent: the level-5 leg reproduces the level-3 leg’s Ψ_4 envelope to 0.005% of peak at both consumer spheres, and its scalar flux to 0.005%, which is the convergence test the spiral’s waveform had otherwise been missing. The level-5 leg restarts from the level-3 checkpoint at $t = 36$, so this bounds the effect of adding two refinement levels to an existing solution rather than of evolving from $t = 0$ at each resolution. The diagnostics are provably non-invasive: two different binaries evolving the same spiral checkpoint produce byte-for-byte identical physics streams over 2384 shared

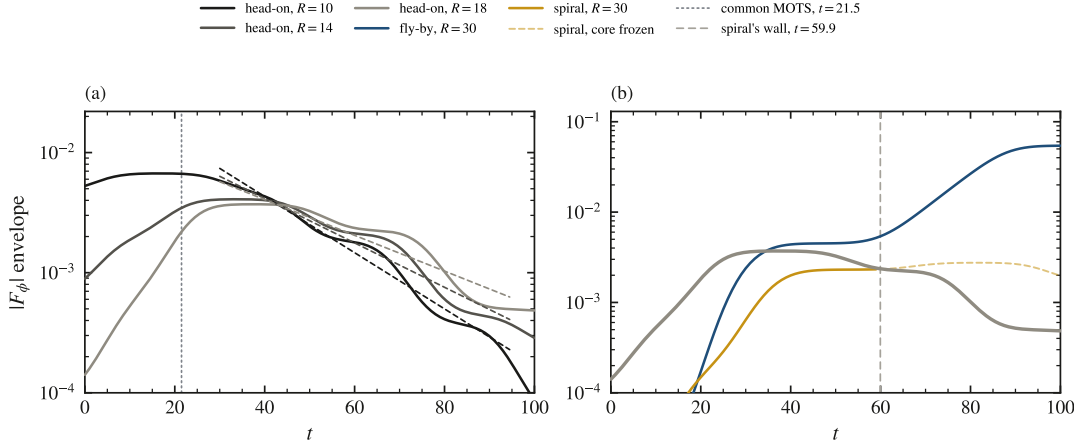


FIG. 14. A horizon, and only a horizon, shuts the ghost dipole off. Each curve is the *envelope* of $|F_\phi|$ —a running maximum over 25 units, one period of the dipole’s own oscillation, smoothed in log amplitude—because the raw flux changes sign every ~ 20 units and its zeros hide the amplitude on a log axis. (a) The head-on arm (level 5 from $t = 0$, scalar stream on) at $R = 10/14/18$. The common MOTS closes at $t = 21.5$ (dotted rule); the remaining hair leaves as one $\ell = 1$ pulse, cresting later at each sphere, and then decays exponentially—dashed, log-linear fits over $t = 30\text{--}95$ with $\tau = 19/23/28$. This is the horizon-mass settling time of Fig. 5(g), $\tau = 19.4$. The rise before the rule is the two open mouths’ static hair superposing, canonically ingoing and inside the near zone; only the post-horizon window is integrated. (b) The control, each encounter on its outermost sphere—head-on $R = 18$, fly-by and spiral $R = 30$, so the colours are those of (a): the fly-by (no horizon, no merger) grows to the end of its record, and the $p = 0.12$ spiral (merges, no MOTS on any scan) is still climbing at its $t = 59.9$ wall (dashed rule). Past that wall the spiral is the frozen-core arm (dashed, half weight): its exterior agrees with the free arm to 0.1% over their overlap, but a frozen source region carries no claim about its own decay.

output rows—which also certifies bit-reproducibility on this hardware. The time step is bracketed (dt multiplier 0.05 is a different solution from $t \approx 33$; 0.1 dies at $t = 16$; 0.02 is used throughout), grid independence after the head-on merger is 0.05–0.33% of peak across three restarted arms, and the placement systematic of Sec. VC is subtracted by construction. The Helfer superposition correction is rejected on measurement: better initial data, worse evolution. It trades the constraint defect at the throats for one in the fade zone between them, and the orbit then stalls at separation 4.7 instead of plunging—at levels 3, 4 and 5 alike, so converged rather than under-resolved, and worse (5.1) when the window is halved; on a fly-by, where nothing merges, the corrected arm’s Hamiltonian norm grows $3.2 \times 10^{-3} \rightarrow 1.6 \times 10^{-2}$ in seven units against a flat 2.2×10^{-3} for its plain twin. These arms do not die—they stall, and were stopped by hand—so production superposes plainly and accepts the initial mouth-size excess of Sec. VC. Constraint-solved data was attempted and is not used either: a *GRTresna* solve reproduces the drainhole once the outer condition on ψ_{reg} is Robin rather than unity (ψ error 4.9% $\rightarrow$ 0.7%), but the bridge that carries the solved grid into the evolution loses the throat, dying at $t = 5.3$ —so every binary here is superposed, with the defect of Sec. IIC declared rather than removed. Sec. XIA shows that removing it would buy a fifth of an orbit. Matter damping shapes nothing it was hoped to shape. Finally, the late Ψ_4 floor growth of the single-throat family is sphere-local, not a boundary reflection: doubling the box leaves the onset at

$t \simeq 41$, and in the spherical control—which cannot radiate a quadrupole—the $R = 14$ sphere climbs four orders of magnitude while the same run’s $R = 30$ sphere never leaves 1.1×10^{-3} . Every wave window in this paper is therefore gated per sphere, ahead of its own onset.

X. PATTERN MATCHING AGAINST DETECTOR DATA

The simulated waveforms are scale-free: with the throat mass M as the unit, $r\Psi_4$ rescales to detector strain by $h \propto M/D$ and $f \propto 1/M$, so one numerical waveform covers every candidate mass at once. This section confronts the rescaled (2, 2) and (2, 0) waveforms with the LIGO–Virgo–KAGRA data products on four fronts: their fitting factor against the template families the catalogues are built from [30], a matched-filter search of the open data itself [26], the ringdown, and the unmodelled searches.

One pipeline produces every number in this section, and it is validated before it is believed. The Ψ_4 records are integrated to strain by fixed-frequency integration [27], with the divisor clamped below a per-arm corner—the lowest instantaneous frequency the record resolves, never below one cycle per record—without which the fly-by, whose frequency genuinely falls below its own $1/T$, is reconstructed a hundred times too loud. The residual pedestal $|h|(\text{ends})/|h|(\text{peak})$ is 0.02–0.07 for the collapsing throat and the vacuum twin, but 0.31–0.42 for

the head-on, spiral and fly-by, whose low-frequency strain is therefore an upper bound and not a measurement; each template is filtered only from its own corner. The end-to-end test that can fail is the campaign’s vacuum binary-black-hole twin, a black-hole merger evolved with the same code, grid and extraction: through this pipeline it matches IMRPhenomD at 0.902–0.938 over 60–300 $M_\odot$ and recovers its total mass to 0–5% at 60–200 $M_\odot$, the missing few per cent accounted for by its eccentricity, its 75 M record and its finite extraction radius. Every fitting factor below is read against that 0.90–0.94 ceiling, never against unity.

A. Fitting factor against the binary-black-hole bank

Against a quasi-circular IMRPhenomD bank spanning 20–500 $M_\odot$ in total mass, mass ratios 1–4 and aligned effective spins ± 0.9 , at Advanced LIGO design sensitivity, the channels return fitting factors of 0.49–0.82: collapsing throat 0.54–0.79, head-on 0.49–0.75, spiral 0.54–0.80, fly-by 0.76–0.82. Taken alone those numbers would say a modelled search recovers 12–56% of the optimal detection volume. The control forbids that reading: the vacuum twin—an ordinary black-hole merger—scores 0.52–0.74 on the same bank, inside the wormhole range. What the full-bank fitting factor measures here is not the exoticism of the source but the *length of the record*: a bank template that spends seconds in band climbing its inspiral has most of its power with nothing to align against, and the recovered masses say the same (0.11–1.84 of the true value; the control’s own at 0.11–0.53). No claim that a wormhole merger would be missed by the modelled searches follows from these records—and none can, because the shortness is the physics rather than a truncation: this matter model cannot produce an inspiral at all (Sec. XI A), so the transient is the whole of the wormhole-phase signal, and what a modelled search would meet at late times is the chirp of whatever the throats convert into, outside these simulations.

What the records do settle is morphology at fixed duration. Cutting each bank template to the signal’s own window removes the duration deficit and leaves the shape: the channels return 0.82–0.98 against the control’s 0.94–0.95—at this duration, burst morphology alone does not separate a drainhole channel from a binary black hole. The discriminator that does is the frequency track, Sec. X C.

B. A matched-filter search of O3b open data

The rescaled waveforms were also run as an actual search of public Advanced LIGO strain [29]. The bank is the five channels times a mass ladder placed so that neighbouring rungs match at 0.97 against the live noise curve—127 templates, total mass the only bank dimension, because the records are scale-free—and the pipeline

is the standard one [26, 28]: matched filter, Allen’s χ^2 bin test, re-weighted SNR, Hanford–Livingston coincidence, background from 5000 time slides. Its sensitivity is measured, not assumed: all 15 injections into conditioned O3b strain at optimal SNR 20 are recovered at 93–139% of optimal (median 103%), within 3 ms, with χ_r^2 near unity. The one weakness is the bank’s shortest template (5 ms, four χ^2 bins), where the veto demotes SNR 27.8 to 14.2—a cost in sensitivity at the lightest fly-by rungs, not in false triggers.

Searching 2.26 h of coincident CBC_CAT3 science time yields *no candidate at zero lag*. The background—13 920 accidental coincidences over 471 days of slid livetime, loudest 60.4—caps the smallest reportable false-alarm rate at one per 471 days; its loudest event is an SNR 758 Hanford glitch that the veto re-weights to 60.4, raising the bar rather than entering the foreground.

The useful output of a null result is reach. At single-detector SNR 8 and optimal orientation, the median horizon over each channel’s mass ladder is 4.8 Gpc (spiral), 2.5 (fly-by), 1.8 (head-on), 1.3 (vacuum twin) and 0.15 (collapsing throat); divide by the standard sky-and-orientation factor $\langle \text{SNR} \rangle^{-1} = 2.26$ (numerically close to the coincident livetime above, and unrelated to it) for the all-sky average, and read the three pedestal-limited channels as upper bounds. The ordering is *not* the energies’ ordering—the spiral reaches furthest although the fly-by radiates $3.4\times$ more—and the reason is that reach is set by where a channel’s power lands in the noise curve, not by how much of it there is: each channel is scanned over its own mass ladder, which moves its spectrum bodily in frequency, and the fly-by’s is the one whose low-frequency content is most nearly all pedestal (Sec. XI B). What survives is the weaker statement the null result actually supports: the horizonless encounters are the ones a detector reaches furthest for, both of them beyond the vacuum control.

C. Ringdown discriminators

Where the burst’s shape does not separate these channels, the way its frequency *moves* does. The reference is fixed by the control: the vacuum twin’s peak sits at $fM = 0.065$, the textbook equal-mass merger value, and its instantaneous frequency climbs $299 \rightarrow 473$ Hz along the Newtonian point-mass track at $30 M_\odot$ before peeling off at merger, heading toward the $\ell = m = 2$ Kerr quasi-normal mode of the equal-mass remnant at $fM = 0.0885$ (599 Hz). That is what a binary black hole does: monotone rise, then a fixed ringdown frequency.

No drainhole channel does it (Fig. 10(c)). The head-on’s signal is an $(\ell, m) = (2, 0)$ bell, and the Kerr reference for it is the axisymmetric $(2, 0, 0)$ mode rather than the $(2, 2, 0)$ track quoted for the twin; against either, its frequency *falls*, $486 \rightarrow 342$ Hz, over a sequence of swings of lengthening period (~ 31 – 38 in units of

M) with the amplitude dropping only $\times 0.6$ – 0.8 per half-swing—a slowly decaying oscillation, not a ringdown, and it falls where a remnant’s would sit fixed. Mass loss cannot rescue the fit, and this is the sharpest form of the discriminator: over the same window the remnant’s Misner–Sharp mass *also* falls, $2.99 \rightarrow 2.17$ (Sec. VIA), and a black hole’s quasi-normal frequency scales as $1/M$, so a shrinking remnant must ring *faster*—by $\times 1.38$ here. The measurement is $\times 0.70$. The frequency drifts the wrong way by a factor of two against its own mass track, which no choice of remnant parameters can absorb. The fly-by falls further and faster, $227 \rightarrow 115$ Hz, as the pair separates; this one is expected, because no horizon ever forms and there is no remnant to ring. The collapsing throat rings at a single frequency, 335 Hz at its peak ($fM = 0.050$, the $20.6 M$ period of Sec. VIII A), whose period is 21% *short* of the Schwarzschild fundamental for its Misner–Sharp mass $M_{\text{MS}} = 1.56$ ($26 M$)—it rings fast for its mass, at 0.55 of the equal-mass Kerr mode.

The practical statement is therefore narrow and defensible. A ringdown-only analysis that fits a damped sinusoid and asks whether its frequency and decay lie on the Kerr (M_f, a_f) track would reject every one of these channels—the head-on because its frequency drifts downward through the fit window instead of holding, the throat because it rings 27% fast for its mass, the fly-by because it is not a ringdown at all. Equivalently, a catalogue event whose post-merger frequency is measured to hold within a few per cent at the Kerr value excludes the head-on channel at that mass. Two limits travel with this. The spiral’s late record is transported by a frozen core from $t = 57$ (Sec. VIII C) and its frequency climbs $151 \rightarrow 679$ Hz without settling, so this campaign has not measured a spiral ringdown and quotes none. And the head-on’s bell is followed only over four swings on a record that ends at $t = 100$, so its decay constant is bounded, not fitted.

D. Burst searches and the stochastic background

An unmodelled search is the right instrument for these signals: a burst pipeline asks only for excess power, coherent between detectors, in a time–frequency region, so the missing inspiral—the feature that costs the modelled bank most of its match, and which this source class cannot supply—costs it nothing. The channels are short and loud in exactly its regime: 6–125 ms across the bank, the energy concentrated within a factor of three in frequency.

The clean, detector-independent deliverable promised in Sec. I is the energy per event in throat-mass units, and this campaign measures it in the dominant multipole, with the spread over extraction spheres as its error bar: fly-by $7.4 \times 10^{-2} M$ (43%), spiral $2.2 \times 10^{-2} M$ (21%), head-on $3.3 \times 10^{-3} M$ (13%), collapsing throat $3.2 \times 10^{-5} M$ (18%), against the equal-parameter vacuum binary’s $2.6 \times 10^{-3} M$ (5%). Since Ω_{GW} from a population scales as the product of the comoving event

density and the energy released per event, these numbers convert any bound on Ω_{GW} —the observing-run limits, or the nucleosynthesis cap that holds outside every detector band [24, 25]—into a bound on the comoving abundance of primordial drainhole encounters, one that is a factor of $\simeq 70$ tighter for the horizonless fly-by than for an equal-mass black-hole binary of the same parameters, and 29 tighter than for a vacuum *merger* of the same masses and separation. Three caveats travel with it. The ghost scalar must exist at the epoch in question to hold the newborn throats open. The fly-by’s 43% sphere spread is the widest in the campaign, because the expansion of its open mouths is near-zone content that the outer spheres see less of. And these are gravitational energies only: the scalar channel carries 2.3 times as much again, in the opposite sign, so a bound derived from Ω_{GW} constrains a population through the smaller of a merger’s two channels—conservatively for the abundance, but it means the quantity a detector limits is not the quantity these events mostly emit (Sec. VIII F).

XI. DISCUSSION

The four fates promised in Sec. I are all realised, each with a measured selection rule: the relative scalar sign separates scattering from merger (Sec. V), the seed’s sign separates collapse from inflation (Sec. IV), and the orbital momentum separates fusing, capture and escape at 50–70% of circular (Sec. VII). What a foam-born population radiates is bursts—the energies of Sec. XD convert any Ω_{GW} bound into an abundance bound—and the budget below shows why no member of the population chirps.

A. No wormhole inspiral

A drainhole binary cannot radiate an inspiral, and the obstruction is a logarithm. Three independent measurements fix the e-fold time of the mouths’ unstable mode at $\tau \simeq 4.4$ code units: an exponential fit to the fly-by’s mouth growth (back-extrapolated seed 7×10^{-4} , the superposition defect) gives 4.39; the eight-unit shift in departure time between refinement levels 3 and 4, divided by $\ln 16$, the fourth-order truncation ratio, gives ~ 2.9 ; and the 11–12-unit delay per decade of declared kick (Sec. IV C) gives ~ 5.0 . In proper time at the minimal surface, where the static lapse is $\alpha_{\text{th}} = e^{u(r_t)} = 0.575$, that is 1.7–2.9, i.e. $T = \tau \alpha_{\text{th}} / R_\star = 0.43$ – 0.74 . These are the *driven* throats of a binary, and they run faster than the isolated mode of Sec. IV B ($T = 0.87$) rather than bracketing it; the comparison with Ref. [14] is made there, on the isolated throat its perturbation theory describes, and is not repeated here.

The budget follows. A quasi-circular pair at separation d needs $t_{\text{insp}} = \frac{5}{256}(d/M)^4 M/\eta$ to merge by radiation reaction (M the total mass, $\eta = 1/4$): 202 units from the innermost stable circular orbit $d/M = 6$ and 640

units from $d/M = 8$ –1.1 and 2.25 orbits, counting orbits at the initial separation ($T_{\text{orb}} = 2\pi\sqrt{d^3/M}$), which understates the cycles actually completed and so states the budget conservatively. The instability is thus handed $t_{\text{insp}}/\tau = 46$ e-folds at the ISCO and 146 for a two-orbit inspiral, so the mouths arrive at merger unchanged only if the initial perturbation is below $e^{-46} \simeq 10^{-20}$, respectively 10^{-63} . Turned around, the lifetime is $\tau \ln(1/\varepsilon)$: seeds of 10^{-6} , 10^{-10} , 10^{-15} buy 0.33, 0.55, 0.82 of one orbit—not one, for any nameable seed. The asymmetry is the whole argument: the orbit’s demand grows as $(d/M)^4$ while the throat’s endurance grows as $\ln(1/\varepsilon)$, so no finer grid, quieter data or better solver closes the gap—constraint-solved initial data, cutting the seed from the superposition defect to the truncation floor, is worth $\tau \ln(2 \times 10^3) \simeq 33$ units, a fifth of one orbit. A companion only sharpens the verdict: a tidal field drives the mode, shortening its e-fold time $5.9 \rightarrow 4.4 \rightarrow 3.7$ from the isolated throat to the fly-by to the merging arm, a 25–37% tightening of the clock the budget is spent against (Sec. VII A). That the budget’s τ is a property of the throat and not of one particular encounter is measured rather than assumed: the arm that merges and the arm that misses, sharing an initial separation and their throats, return 3.70 and 4.39 from the same fitted window (Fig. 6).

Two corrections to the budget’s own machinery both point the same way, and neither is small. The inspiral time above is the vacuum, quadrupole, quasi-circular result, and this binary satisfies none of its premises. Its constituents carry opposite scalar charges, so their leading radiative moment is a *dipole*, not a quadrupole, and the ratio of scalar to gravitational emission measured here is 2.3 on the same sphere (Sec. VIII F)—so the dominant channel is one the formula does not contain. And that channel’s energy is negative, so it pumps the orbit rather than draining it. Under the ordinary sign, a radiation reaction a factor of a few stronger than assumed only shortens t_{insp} and hands the instability fewer e-folds while it still needs $\ln(1/\varepsilon) \gtrsim 46$: the verdict is unchanged and reached sooner. Under the sign this matter actually has, the orbit is driven outward by its own emission and a quasi-circular decay does not exist to be computed. A circularised wormhole inspiral therefore fails twice over—once on the throat’s clock and once on the orbit’s—and the arithmetic above, which uses the vacuum formula, is the conservative of the two arguments rather than the fragile one.

The wormhole-specific signal is therefore a short transient: within a few tens of M each mouth has collapsed—forming a horizon at $t = 11$ and 25 for kicks of 10^{-2} and 10^{-3} —or inflated, measured to $\times 7.8$ in the fly-by, and the plunge, burst and conversion of Secs. VI–VIII are the whole of what this source class emits *as wormholes*. Three qualifications are mandatory. The statement covers the matter model the González-Guzmán-Sarbach theorem covers, a minimally coupled ghost scalar; throats held open by rotation, additional fields, thin shells or

higher-curvature terms are untouched by it. The system converts rather than falls silent, so “no inspiral” means no *wormhole* inspiral: any long chirp belongs to the objects the throats become. And the seeds measured here are numerical—superposition defect and truncation noise—which the argument tolerates only because its dependence on the seed is logarithmic; no physical environment supplies a perturbation below 10^{-20} .

B. What is not established

Collected here so that no claim above has to carry its own retraction.

1. **No horizon is measured for the spiral merger.** No trapped surface exists anywhere on its measured record, and the curvature wall that kills every grid rides the still-open throat; a $16\times$ refinement ladder saturates without reaching the merger’s end (Sec. VII).
2. **No spiral ringdown is measured.** The late record is transported by a core frozen from $t = 57$, so its swings are propagation, not a remnant ringing; the instantaneous frequency climbs $151 \rightarrow 679$ Hz without settling (Sec. X C).
3. **The converted pair’s own signal is not simulated.** After the wormhole transient the constituents are black holes or inflating runaways (Sec. XI A); the chirp of such a converted binary—what a modelled search would meet at late times—is outside these records, which bound only the wormhole-phase morphology (Sec. X A).
4. **The merging spiral’s mouths are measured only to $t = 50$ and only at level 3.** The instrumented re-run (Fig. 6) gives the mouth sizes through the plunge, $+12.2\%$ by $t = 28$, but its per-mouth reading ends where the two scan spheres meet, and the production record itself still carries no areal-radius stream: how far the throats had gone at the level-5 death is inferred from the clock, not read off.
5. **The low-frequency strain of three channels is an upper bound.** Head-on, spiral and fly-by reconstruct with 31–42% of their peak strain still present at the record’s ends, so their sub-corner content is integration pedestal rather than measurement (Sec. X); the collapsing throat and the vacuum control, at 2–7%, do not have this problem.
6. **The late constraint growth on both single-throat branches from $t \approx 75$ is unexplained,** as is the grid-scale wobble that appears on the outermost head-on sphere after $t \approx 80$ (Sec. IX).

7. **The energy balance is not closed.** Sec. VIII F establishes that the scalar channel is dipole-dominated, of order twice the gravitational energy and negative—but not that $\dot{M}_{\text{ADM}} = -F_{\text{GW}} - F_{\phi}$ holds, because no surface-integral ADM mass was recorded on any arm. The scalar flux itself is a coordinate flux with no lapse or shift factors, quoted with a 30% sphere spread, and one of the five channels—the single throat—still has no scalar measurement at all: its arms predate the stream and their plotfiles are not retained, so closing it requires a re-run, not a re-read. The head-on was closed that way (Sec. VIII G), and only its post-horizon window is usable: before the horizon the flux at the inner sphere runs ingoing, the two open mouths’ static hair rather than radiation. For a system whose novelty is negative-energy matter this is the largest single gap in the campaign.
8. **The horizon scan is star-shaped about a chosen centre.** Every horizon statement—formation, absence, radius, mass—comes from coordinate spheres rather than a deformable finder (Sec. III C). The shape systematic therefore attaches not only to the quoted masses but to the *absence* claims that carry the most weight: “born common or not at all” (Sec. VI A) and “no trapped surface anywhere on the spiral’s record” (Sec. VII) are statements about star-shaped surfaces about the tracked pits and the box centre, not about all surfaces.
9. **The comparison with González–Guzmán–Sarbach is a rate comparison, not a parameter-matched one.** The map from this paper’s (a, m) drainhole to that work’s perturbation parameters has not been verified term by term, so the agreement quoted in Sec. IV B is $\sim 15\%$ against the published growth times and no finer statement is supported.

XII. REPRODUCIBILITY

Every evolution in Table I is one line in a registry shipped with the results pack—its single knob against a named reference, its outcome, its caveats—and every

figure in this paper is regenerated by a script in the same pack from the archived diagnostic streams. The evolutions are bit-reproducible on the production hardware: two builds differing only in a passive diagnostic reproduce 2384 shared output rows byte for byte (Sec. IX). The code is `GRTEclyn` [39]; the drainhole initial data, the seeds of Sec. IID, the orientation-corrected surface scan and the parameter file of every run accompany the registry.

XIII. COMPUTATIONAL COST

Every evolution ran on one NVIDIA H100 80 GB device of a four-H100 node with two 32-core Xeon Platinum 8462Y+ processors—up to four arms concurrently while the cards were free—with `GRTEclyn` built CUDA-native against AMReX 26.02. At $\Delta t = 0.02 \Delta x$ a production arm ($L = 64$, 128^3 base, three refinement levels) advances 12–18 code units per hour, so a $t = 100$ evolution costs 6–8 h. Depth is the expense: the level-5 arms run at 2–4 units per hour—28.2 h for the head-on’s seamless $t = 0 \rightarrow 100$ (the level-3 $\rightarrow 5$ chain it replaced cost 18.6 h for its $t = 22 \rightarrow 100$ leg alone), 46.7 h for the fly-by’s full $t = 0 \rightarrow 100$ on the doubled box, which peaks at 58 of the 80 GB with 26-GB checkpoints against a rolling keep-list—and the production spiral’s three-stage chain (Sec. VII C) costs 23.5 h end to end. The 121 evolutions this paper draws on total 708 GPU-hours, recomputed from each run’s own logged speed and time span by a script shipped with the results pack; the archived shake-down stage adds 7 h more, for 716 h of packed evolution in all. Earlier drafts quoted a higher figure, which charged the campaign twice for ten runs the prune had left a second copy of; the script now drops copies and dead launches by name rather than counting directories.

ACKNOWLEDGMENTS

This work was supported by Gravity Frontiers (<https://www.gravityfrontiers.org/en>), which funded the underlying research and the development of the numerical-simulation software. The author extends sincere gratitude to Ilya Nachevsky for generously providing the high-performance computing resources essential to this work, and acknowledges the GRTL Collaboration for `GRChombo`, `GRtresa`, and `GRTEclyn`.

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- [1] J. A. Wheeler, “Geons,” *Phys. Rev.* **97**, 511 (1955).
 - [2] J. A. Wheeler, “On the nature of quantum geometrodynamics,” *Ann. Phys. (N.Y.)* **2**, 604 (1957).
 - [3] C. W. Misner and J. A. Wheeler, “Classical physics as geometry,” *Ann. Phys. (N.Y.)* **2**, 525 (1957).
 - [4] T. A. Roman, “Inflating Lorentzian wormholes,” *Phys. Rev. D* **47**, 1370 (1993).
 - [5] J. Garriga, A. Vilenkin, and J. Zhang, “Black holes and the multiverse,” *J. Cosmol. Astropart. Phys.* **02**, 064

(2016).

- [6] H. Deng, J. Garriga, and A. Vilenkin, “Primordial black hole and wormhole formation by domain walls,” *J. Cosmol. Astropart. Phys.* **04**, 050 (2017).
- [7] A. A. Kirillov and E. P. Savelova, “Dark matter from a gas of wormholes,” *Phys. Lett. B* **660**, 93 (2008).
- [8] A. A. Kirillov and E. P. Savelova, “Density perturbations in a gas of wormholes,” *Mon. Not. R. Astron. Soc.* **412**, 1710 (2011).

- [9] H. G. Ellis, “Ether flow through a drainhole: A particle model in general relativity,” *J. Math. Phys.* **14**, 104 (1973).
- [10] K. A. Bronnikov, “Scalar-tensor theory and scalar charge,” *Acta Phys. Polon. B* **4**, 251 (1973).
- [11] M. S. Morris and K. S. Thorne, “Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity,” *Am. J. Phys.* **56**, 395 (1988).
- [12] M. Visser, *Lorentzian Wormholes: From Einstein to Hawking* (AIP Press, Woodbury, NY, 1995).
- [13] H. Shinkai and S. A. Hayward, “Fate of the first traversible wormhole: Black-hole collapse or inflationary expansion,” *Phys. Rev. D* **66**, 044005 (2002).
- [14] J. A. González, F. S. Guzmán, and O. Sarbach, “Instability of wormholes supported by a ghost scalar field. II. Nonlinear evolution,” *Class. Quantum Grav.* **26**, 015011 (2009); see also **26**, 015010 (2009).
- [15] R. Geroch, “Topology in general relativity,” *J. Math. Phys.* **8**, 782 (1967).
- [16] F. J. Tipler, “Singularities and causality violation,” *Ann. Phys. (N.Y.)* **108**, 1 (1977).
- [17] S. W. Hawking, “Wormholes in spacetime,” *Phys. Rev. D* **37**, 904 (1988).
- [18] Ya. B. Zel’dovich and I. D. Novikov, “The hypothesis of cores retarded during expansion and the hot cosmological model,” *Sov. Astron.* **10**, 602 (1967).
- [19] B. J. Carr and S. W. Hawking, “Black holes in the early Universe,” *Mon. Not. R. Astron. Soc.* **168**, 399 (1974).
- [20] B. Carr, K. Kohri, Y. Sendouda, and J. Yokoyama, “Constraints on primordial black holes,” *Rep. Prog. Phys.* **84**, 116902 (2021).
- [21] C. Caprini and D. G. Figueroa, “Cosmological backgrounds of gravitational waves,” *Class. Quantum Grav.* **35**, 163001 (2018).
- [22] G. Agazie *et al.* (NANOGrav Collaboration), “The NANOGrav 15 yr data set: Evidence for a gravitational-wave background,” *Astrophys. J. Lett.* **951**, L8 (2023).
- [23] P. Amaro-Seoane *et al.*, “Laser Interferometer Space Antenna,” arXiv:1702.00786 [astro-ph.IM] (2017).
- [24] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Upper limits on the isotropic gravitational-wave background from Advanced LIGO and Advanced Virgo’s third observing run,” *Phys. Rev. D* **104**, 022004 (2021).
- [25] L. Pagano, L. Salvati, and A. Melchiorri, “New constraints on primordial gravitational waves from Planck 2015,” *Phys. Lett. B* **760**, 823 (2016).
- [26] B. Allen, W. G. Anderson, P. R. Brady, D. A. Brown, and J. D. E. Creighton, “FINDCHIRP: An algorithm for detection of gravitational waves from inspiraling compact binaries,” *Phys. Rev. D* **85**, 122006 (2012).
- [27] C. Reisswig and D. Pollney, “Notes on the integration of numerical relativity waveforms,” *Class. Quantum Grav.* **28**, 195015 (2011).
- [28] S. A. Usman *et al.*, “The PyCBC search for gravitational waves from compact binary coalescence,” *Class. Quantum Grav.* **33**, 215004 (2016).
- [29] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Open data from the third observing run of LIGO, Virgo, KAGRA and GEO,” *Astrophys. J. Suppl.* **267**, 29 (2023).
- [30] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “GWTC-3: Compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run,” *Phys. Rev. X* **13**, 041039 (2023).
- [31] C. W. Misner and D. H. Sharp, “Relativistic equations for adiabatic, spherically symmetric gravitational collapse,” *Phys. Rev.* **136**, B571 (1964).
- [32] D. Hochberg and M. Visser, “Dynamic wormholes, anti-trapped surfaces, and energy conditions,” *Phys. Rev. D* **58**, 044021 (1998).
- [33] J. M. Bowen and J. W. York, “Time-asymmetric initial data for black holes and black-hole collisions,” *Phys. Rev. D* **21**, 2047 (1980).
- [34] T. Helfer, U. Sperhake, R. Croft, M. Radia, B.-X. Ge, and E. A. Lim, “Malaise and remedy of binary boson-star initial data,” *Class. Quantum Grav.* **39**, 074001 (2022).
- [35] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, “Conformal and covariant formulation of the Z4 system with constraint-violating solutions,” *Phys. Rev. D* **85**, 064040 (2012).
- [36] C. Bona, J. Massó, E. Seidel, and J. Stela, “New formalism for numerical relativity,” *Phys. Rev. Lett.* **75**, 600 (1995).
- [37] M. Campanelli, C. O. Lousto, P. Marronetti, and Y. Zlochower, “Accurate evolutions of orbiting black-hole binaries without excision,” *Phys. Rev. Lett.* **96**, 111101 (2006).
- [38] J. G. Baker, J. Centrella, D.-I. Choi, M. Koppitz, and J. van Meter, “Gravitational-wave extraction from an inspiraling configuration of merging black holes,” *Phys. Rev. Lett.* **96**, 111102 (2006).
- [39] GRTL Collaboration, *GRTEclyn*, <https://github.com/GRTLCollaboration/GRTEclyn>.
- [40] W. Zhang *et al.*, “AMReX: a framework for block-structured adaptive mesh refinement,” *J. Open Source Softw.* **4**, 1370 (2019).
- [41] T. Andrade *et al.*, “GRChombo: an adaptable numerical relativity code for fundamental physics,” *J. Open Source Softw.* **6**, 3703 (2021).
- [42] N. M. Shirokov, “Wormhole dynamics: nonlinear collapse and gravitational-wave emission,” arXiv:2604.00071 [gr-qc] (2026).
- [43] N. M. Shirokov, “The Bondi dipole in full numerical relativity: a self-accelerating positive-negative mass binary,” submitted (2026).